

The Schooling Lag

Rank Persistence and the Measurement of Intergenerational Educational Mobility

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Abstract

The two workhorses of the intergenerational mobility literature—the intergenerational elasticity (IGE) and the rank-rank regression—both fail when applied to educational data. The IGE conflates genuine changes in positional persistence with the mechanical compression of the schooling distribution across generations (Solon, 1999; Hertz et al., 2007). The rank-rank regression, designed for continuous income, breaks down with discrete education: massive ties corrupt the estimand and standard inference is invalid (Chetverikov and Wilhelm, 2024; Ahsan et al., 2024). This paper proposes a two-step estimator—*demean by cohort, then rank the residuals*—that resolves both problems simultaneously. The schooling lag—a respondent’s deviation from their cohort-sex mean—transforms a discrete variable with K mass points into a near-continuous one with up to $K \times G$ distinct values, where G is the number of cohort-sex groups. We formalize this within a four-object decomposition that nests the IGE, the demeaned-level slope, the rank-of-levels slope, and the rank-of-lags slope, showing precisely how each transformation removes a distinct source of bias. The same deflation affects sibling correlations: the broadest measure of family effects systematically understates the family share during educational expansion. A second, orthogonal bias arises from the fertility transition: as educated families reduce fertility first, the composition of who contributes sibling pairs shifts, compressing between-family variance independently of any change in family effects. We derive a within-cohort Great Gatsby Curve—the

structural relationship $\omega^2 = \sigma_\varepsilon^2 / (1 - \rho^2)$ between within-cohort educational inequality (ω) and rank persistence (ρ)—which separates access expansion (which moves the IGE but not ρ) from quality equalization (which moves both). Applied to nineteen waves of Latinobarómetro spanning 1998–2023 across eighteen Latin American countries, the schooling lag shows that approximately three-quarters of the observed decline in the population-weighted IGE is a distributional artifact from falling σ_c / σ_p . The within-cohort correlation ρ declined modestly—from 0.48 to 0.45 over 25 years—while β_A declined more than three times as fast. The standard IGE overstates mobility progress by a factor of three.

JEL Classification: I24 · J62 · D63 · C21 · O15

Keywords: schooling lag; intergenerational mobility; rank-rank correlation; demean-then-rank; Great Gatsby Curve; educational expansion; Latin America

1 Introduction

How should we measure whether the family one is born into predicts where one stands in the educational distribution of one’s own generation? The intergenerational mobility literature offers two standard answers, and both fail for education.

The first answer is the intergenerational elasticity (IGE): regress child’s years of schooling on parent’s years. [Solon \(1999\)](#) showed that this slope equals the intergenerational correlation r multiplied by the ratio of cross-sectional standard deviations σ_c/σ_p . When educational expansion compresses the child distribution relative to the parent—as it has done throughout the developing world since 1960—the IGE falls mechanically, even when rank persistence is unchanged. [Hertz et al. \(2007\)](#) documented this globally: across forty-two countries and fifty years, the regression coefficient fell from 0.80 to 0.50 while the correlation held steady at 0.43. [Neidhöfer et al. \(2018\)](#) found the same for Latin America, noting that Pearson and Spearman correlations were “relatively stable around 0.5 over the entire period” even as the IGE declined. [Torche \(2019\)](#) stated the implication directly: the decline “is entirely driven by changes in the distribution of schooling.”

The second answer is the rank-rank regression: convert both generations’ schooling to within-cohort percentile ranks and regress child rank on parent rank. Introduced by [Dahl and DeLeire \(2008\)](#) and popularized by [Chetty et al. \(2014\)](#), this approach kills the σ -ratio by forcing both marginals to uniform, making the slope equal to the Spearman rank correlation. For continuous income observed in large administrative datasets, the rank-rank specification is demonstrably superior: it is robust to lifecycle bias, handles zeros naturally, produces strikingly linear relationships, and enables clean decompositions of absolute and relative mobility ([Chetty et al., 2014](#); [Nyblom and Stuhler, 2016](#)).

But education is not income. Years of schooling is a discrete variable taking at most 17 values in the best data and as few as 7 in typical survey recodes. [Chetverikov and Wilhelm \(2024\)](#), in a paper conditionally accepted at *Econometrica*, prove that with point masses the rank-rank slope can fall outside $[-1, 1]$, depends on the arbitrary tie-breaking rule,

and cannot be interpreted as a correlation. In their India application with seven education categories, the slope ranges from 0.4 to 1.0 depending on how ties are handled, while the Spearman correlation remains stable at approximately 0.5. [Ahsan et al. \(2024\)](#) show that in a difference-in-differences evaluation of Indonesia’s Inpres school construction—a program with a known large positive effect on schooling ([Duflo, 2001](#))—the standard rank-rank specification *eliminates the treatment effect entirely*, while correctly computed ranks recover it at high significance. The rank-rank regression applied to discrete education does not merely attenuate genuine effects; it can erase them.

The literature has arrived at an impasse. The IGE has the σ -ratio problem. The rank-rank has the tie problem. [Asher et al. \(2024\)](#) propose partial identification bounds as an alternative but sacrifice the single-number comparability that makes the IGE and rank-rank useful for cross-country and cross-time comparison. [Ahsan et al. \(2022\)](#) and [Kunievsky \(2024\)](#) diagnose the problem with increasing precision but offer no constructive resolution.

This paper proposes a way out. The key insight is that the σ -ratio problem and the tie problem share a common source: the compression of the educational distribution across generations. A procedure that targets this source should mitigate both symptoms simultaneously. The procedure is simple: (1) subtract cohort-sex-country means from each individual’s schooling, producing a *schooling lag*; (2) rank the pooled lags within country.

Step 1 is a linear location shift that strips the between-cohort mean trajectory—the dominant component of educational expansion. This removes the mechanical part of the σ -ratio that drives the IGE down during expansion. But Step 1 also does something less obvious: it generates continuity. A discrete variable with K mass points, after demeaning by G cohort-sex groups with distinct means, produces up to $K \times G$ distinct values ([Proposition 2](#)). With 17 education levels and 40 cohort-sex cells, the effective number of mass points rises from 17 to approximately 680. The tie structure that corrupts the rank-of-levels specification is dramatically reduced.

Step 2 ranks these near-continuous lags, standardizing both marginals to approximately

uniform and killing the residual within-cohort variance ratio ω_c/ω_p that survives Step 1 alone. The resulting estimator—the *rank-of-lags slope*—approximates the intergenerational rank correlation ρ without either the σ -ratio contamination of the IGE or the tie-induced pathologies of the standard rank-rank.

We formalize this within a four-object decomposition (Section 2) that nests the IGE (β_A), the demeaned-level slope (β_R), the rank-of-levels slope (β_{rank}), and the rank-of-lags slope (β_{DR}). The framework shows that each object equals the rank correlation ρ scaled by a distinct variance ratio; the four objects form a hierarchy in which each successive transformation peels off one source of bias.

We then derive a within-cohort Great Gatsby Curve (Section 3) that gives the rank-of-lag estimator a structural interpretation. In a Solon-type OLG model, the cohort mean absorbs all public investment that raises schooling uniformly; the schooling lag strips it, leaving only the private transmission advantage δ and luck. The steady-state within-cohort variance satisfies $\omega^2 = \sigma_\epsilon^2 / (1 - \rho^2)$: a structural, causal relationship between within-cohort inequality and rank persistence. This yields a clean educational Gatsby Curve— ω against ρ —that contains no mechanical component, unlike the standard GGC where the inequality measure (σ_c/σ_p or Gini) enters the mobility measure (β_A) by construction. The within-cohort Gatsby Curve separates three policy channels: access expansion (which raises $\bar{s}_g$ but leaves ω and ρ unchanged), quality equalization (which reduces both ω and ρ), and elite capture (which raises both).

The empirical application (Section 6) uses eighteen waves of Latinobarómetro spanning 1998–2023 across eighteen Latin American countries—the longest panel of cross-country educational mobility available for the region. The results are striking. The population-weighted IGE declined from 0.46 to 0.37 (-0.067 per decade, $p < 0.001$), but approximately three-quarters of this decline is artifact: σ_c/σ_p fell from 0.97 to 0.81 over the same period. The within-cohort correlation ρ_{lag} declined modestly—from 0.48 to 0.45 (-0.020 per decade, $p = 0.013$)—and the rank-of-lag slope β_{DR} tracks it closely (-0.017 per decade).

Decomposing, approximately 76% of the IGE decline reflects distributional compression and 24% reflects genuine improvement in positional mobility. The standard measure overstates progress by roughly a factor of three. At the country level, the decomposition reshuffles mobility rankings: Chile, which appears among the most mobile in β_A , ranks in the persistently immobile half under ρ_{lag} ; Mexico and Peru, moderate in β_A , emerge as the region’s most positionally rigid societies.

1.1 Related literature

This paper engages three literatures.

The measurement of intergenerational mobility. The decomposition $\beta = r \cdot \sigma_c / \sigma_p$ originates in [Solon \(1999\)](#), with cross-country implications developed in [Solon \(2002\)](#) and the Becker-Tomes structural interpretation in [Solon \(2004\)](#). [Black and Devereux \(2011\)](#) survey the econometrics comprehensively. [Nyblom and Stuhler \(2016\)](#) show rank-rank has the smallest lifecycle bias; [Nyblom and Stuhler \(2017\)](#) show it is attenuated by measurement error in *both* variables, unlike the IGE. [Nyblom and Stuhler \(2024\)](#), the lead article in the August 2024 *JPE*, prove that the IGE and the rank correlation can respond with *opposite signs* to the same structural change—the definitive demonstration that the choice of measure is not merely a matter of scaling. [Mogstad and Torsvik \(2023\)](#) show that rank-rank is not invariant to changes in inequality independent of parental rank, using a gender-discrimination counterexample where random wage gaps between men and women dilute the rank-rank correlation, making a discriminatory society *appear* more mobile. [Deutscher and Mazumder \(2023\)](#) compare nineteen measures using Australian administrative data. [Kunievsky \(2024\)](#) decomposes the rank-rank coefficient via Yitzhaki’s theorem, finding it captures copula structure but “lacks insights into the underlying mechanisms.”

The discreteness critique is developed in [Chetverikov and Wilhelm \(2024\)](#), [Ahsan et al. \(2022\)](#), [Ahsan et al. \(2024\)](#), and [Asher et al. \(2024\)](#). We show that two distinct biases

affect sibling correlations during educational expansion: between-cohort variance inflates the ICC denominator (Proposition 3), and the fertility transition changes which families contribute sibling pairs, compressing the ICC numerator (Proposition 4). The schooling lag corrects the first but not the second; only reweighting to a fixed family-composition benchmark addresses both. Our demean-then-rank estimator connects to the jittering approach of Machado and Santos Silva (2005), who add uniform noise to count data for quantile regression; the copula-with-covariates insight of Trivedi and Zimmer (2017), who show that continuous conditioning variables restore copula identification for discrete outcomes; and the concordance-measure bounds of Denuit and Lambert (2005), who prove Spearman’s ρ is restricted to sub-intervals of $[-1, 1]$ for discrete data.

Sibling correlations. The sibling correlation—the ICC from a random effects model—is the broadest measure of family effects, capturing genetics, parenting, neighborhood, and all other shared channels (Solon, 1999). Mazumder (2008) established REML estimation as the standard; Björklund and Jäntti (2012) decomposed the variance into birth-order and family components. The standard specification controls for birth-year dummies and gender but deliberately excludes birth order, since the latter is a within-family source of inequality that the omnibus ICC is designed to capture (Black et al., 2005). Björklund et al. (2010) showed that sibling correlations decline with age spacing: restricting to siblings within 9 years raises brother correlations by 20%. Three recent large-scale databases document cross-country patterns: Muñoz and Jaqué (2024) compiled sibling correlations for 94 countries, finding that global declines are concentrated in Latin America and North America; Ahsan et al. (2023) estimated sibling correlations for 53 developing countries using 230 DHS waves, reporting the highest regional average for LAC ($\rho_{\text{sib}} = 0.65$) with monotonic declines across birth cohorts; and Anderson et al. (2024) meta-analyzed 18 countries, finding no overall secular trend but a positive Gatsby-type association between childhood inequality and sibling correlations. Two open questions remain: whether the documented LAC decline

reflects genuine equalization or mechanical deflation from distributional compression (Proposition 3), and whether the demographic fertility transition—which changed which families contribute sibling pairs—independently biases the ICC (Proposition 4). Neither Muñoz and Jaqué (2024) nor Ahsan et al. (2023) addresses either channel.

Intergenerational educational mobility in Latin America. Behrman et al. (2001) pioneered the cross-country comparison; Hertz et al. (2007) placed LAC in global context, finding the seven most persistent countries worldwide were all Latin American. Neidhöfer et al. (2018) constructed the comprehensive 18-country database, documenting the inverted-U in absolute mobility and its association with inequality. Neidhöfer (2019) tested the within-LAC Great Gatsby Curve, finding childhood inequality negatively and significantly associated with adult mobility. van der Weide et al. (2024) extended coverage to 153 countries in the World Bank’s Global Database on Intergenerational Mobility. Celhay and Gallegos (2025) studied three-generation mobility in six LAC countries, finding multigenerational persistence 77% higher than the Becker-Tomes prediction. Brunori et al. (2025) linked intergenerational mobility to inequality of opportunity. Torche (2014) and Torche (2019) provide authoritative reviews.

The Great Gatsby Curve. Corak (2013) established the cross-country Gini–IGE correlation; Neidhöfer (2019) confirmed it within LAC. Solon (2004) provided the Becker-Tomes structural foundation—the steady-state IGE increases with returns to human capital and decreases with progressivity of public investment. Our within-cohort Gatsby Curve extends this by separating between-cohort public investment (which enters the cohort mean and is stripped by the lag) from within-cohort persistence (which is the ρ the lag recovers).

1.2 Outline

Section 2 presents the four-object decomposition and shows that the same deflation affects sibling correlations (Section 2.5). Section 3 derives the within-cohort Great Gatsby Curve. Section 4 describes the data. Section 5 validates the estimator via Monte Carlo simulation. Section 6 presents the empirical results. Section 7 discusses implications. Section 8 concludes. The Online Appendix contains data construction details, quality filters, decomposition checks, and robustness tests.

2 The Four Objects

2.1 Setup

Individual i in generation $t \in \{p, c\}$ (parent, child) has completed years of schooling $s_i^{(t)} \in \{0, 1, \dots, S\}$. Each individual belongs to a cohort-sex-country group $g(i)$, defined by five-year birth cohort, sex, and country. The census-based mean schooling for group g is $\bar{s}_g$; the within-group CDF is $F_g(\cdot)$. We observe parent-child pairs (s_i^p, s_i^c) from survey data.

The total variance of schooling in generation t decomposes into between-cohort and within-cohort components:

$$\sigma_t^2 = \underbrace{\text{Var}(\bar{s}_g)}_{\sigma_{B,t}^2} + \underbrace{\mathbb{E}[\text{Var}(s^{(t)}|g)]}_{\omega_t^2} \quad (1)$$

Define $\lambda_t \equiv \omega_t/\sigma_t \in (0, 1]$, the *within-cohort share* of total standard deviation. Educational expansion operates primarily through the between-cohort channel (rising $\bar{s}_g$ for later cohorts), so during expansion $\sigma_{B,t}^2$ rises while ω_t^2 changes more slowly.

2.2 Definitions

We apply two transformations—demeaning and ranking—in different orders, generating four mobility estimands.

Definition 1 (Object 1: The Absolute IGE).

$$\beta_A \equiv \frac{\text{Cov}(s^c, s^p)}{\text{Var}(s^p)} = r \cdot \frac{\sigma_c}{\sigma_p} \quad (2)$$

where $r = \text{Corr}(s^c, s^p)$ and $\sigma_t = \text{SD}(s^{(t)})$. This is the standard workhorse of the mobility literature (Solon, 1999; Hertz et al., 2007; Neidhöfer et al., 2018; van der Weide et al., 2024).

Definition 2 (Object 2: The Demeaned-Level Slope). The schooling lag is $\tilde{s}_i^{(t)} \equiv s_i^{(t)} - \bar{s}_{g(i)}^{(t)}$: the deviation of individual i 's schooling from the mean of her cohort-sex-country group, computed from external census data. The demeaned-level slope is:

$$\beta_R \equiv \frac{\text{Cov}(\tilde{s}^c, \tilde{s}^p)}{\text{Var}(\tilde{s}^p)} = \rho_{lag} \cdot \frac{\omega_c}{\omega_p} \quad (3)$$

where $\rho_{lag} = \text{Corr}(\tilde{s}^c, \tilde{s}^p)$ and $\omega_t = \text{SD}(\tilde{s}^{(t)})$.

The schooling lag answers: how many years above or below her own cohort's average did this individual achieve? An individual with 6 years in a cohort averaging 4.3 has a lag of +1.7; the same 6 years in a cohort averaging 10.1 yields −4.1. The lag redefines educational achievement as a positional concept.

Definition 3 (Object 3: The Rank-of-Levels Slope). Let $R_i^{(t)} = F_{g(i)}^{(t)}(s_i^{(t)})$ be the percentile rank of s_i within cohort-sex group $g(i)$ —the Chetty et al. specification applied to education. Then:

$$\beta_{rank} \equiv \frac{\text{Cov}(R^c, R^p)}{\text{Var}(R^p)} \stackrel{\text{continuous}}{=} \rho_{rank} \quad (4)$$

The equality holds when R^c and R^p have equal variances, which requires both marginals to be (approximately) uniform. For continuous income, this is guaranteed by the probability integral

transform. For discrete education with K mass points, the transform produces a step function with at most K distinct values per cohort, and the equality fails ([Chetverikov and Wilhelm, 2024](#)).

Definition 4 (Object 4: The Rank-of-Lags Slope). First compute the schooling lag $\tilde{s}_i^{(t)}$; then define $\tilde{R}_i^{(t)}$ as the percentile rank (or decile) of $\tilde{s}_i^{(t)}$ in the pooled within-country distribution of lags. Then:

$$\beta_{DR} \equiv \frac{\text{Cov}(\tilde{R}^c, \tilde{R}^p)}{\text{Var}(\tilde{R}^p)} \approx \rho_{DR} \quad (5)$$

The approximation holds when the pooled lag distribution is approximately continuous—a condition ensured by the demeaning step ([Proposition 2](#)).

2.3 The nested decomposition

Proposition 1 (Variance-ratio hierarchy). Under the approximation $r \approx \rho_{lag}$ (which holds when cohort means vary smoothly and the transmission parameter does not vary systematically across cohorts):

$$\beta_A \approx \rho \cdot \frac{\sigma_c}{\sigma_p} \quad (6)$$

$$\beta_R \approx \rho \cdot \frac{\omega_c}{\omega_p} \quad (7)$$

$$\beta_{DR} \approx \rho \quad (8)$$

The demeaned-level slope removes the between-cohort component of the σ -ratio; the rank-of-lag slope removes the remaining within-cohort component. Specifically:

$$\frac{\beta_R}{\beta_A} \approx \frac{\lambda_c}{\lambda_p}, \quad \frac{\beta_{DR}}{\beta_R} \approx \frac{\omega_p}{\omega_c}, \quad \frac{\beta_{DR}}{\beta_A} \approx \frac{\sigma_p}{\sigma_c} \quad (9)$$

Proof. The first ratio follows from $\beta_R/\beta_A = (\rho \cdot \omega_c/\omega_p)/(\rho \cdot \sigma_c/\sigma_p) = (\omega_c/\omega_p)/(\sigma_c/\sigma_p) = \lambda_c/\lambda_p$. The second from $\beta_{DR}/\beta_R = \rho/(\rho \cdot \omega_c/\omega_p) = \omega_p/\omega_c$. The third by composition. $\square$

The approximation $r \approx \rho_{lag}$ requires that within-cohort deviations of children be uncorrelated with the cohort means of parents. Formally, $\text{Cov}(\bar{s}_g^c, \bar{s}_g^p) \approx 0$. This holds when cohort means shift smoothly—a condition satisfied empirically, as the mean absolute deviation between β_A and $\hat{\rho}_{lag} \cdot (\sigma_c/\sigma_p)$ is 0.014 across 335 country-wave observations in our data.

Corollary 1 (The inverted-U is mechanical). *If educational expansion generates a declining σ_c/σ_p while measured $\hat{\rho}$ is approximately constant or rising, $\beta_A = \hat{\rho} \cdot \sigma_c/\sigma_p$ can trace an inverted-U trajectory even with constant δ . More precisely, β_A rises when $\Delta\hat{\rho}/\hat{\rho} > |\Delta(\sigma_c/\sigma_p)/(\sigma_c/\sigma_p)|$ and falls when the compression dominates.*

This is relevant because [Neidhöfer \(2019\)](#) documented an inverted-U pattern in Latin American educational mobility, with the IGE rising through 2008 and falling after. Corollary 1 shows this pattern is consistent with constant δ and a boom-bust cycle in σ_c/σ_p —a possibility the simulation in Section 5 confirms.

Remark 1. *Object 3 (β_{rank}) is absent from the hierarchy because its relationship to ρ is contaminated by the tie problem documented by [Chetverikov and Wilhelm \(2024\)](#): the OLS slope equals the Spearman $\hat{\rho}_S$ multiplied by $\text{SD}(R^Y)/\text{SD}(R^X)$, a ratio that departs from unity—potentially far—when the underlying variable has point masses. Object 4 achieves the same goal as Object 3 (killing the variance ratio) without inheriting this pathology, because the demeaning step reduces ties before the ranking step encounters them. The problem worsens during ceiling-approaching expansion: as more individuals pile onto the top education categories, the tie structure degrades further, and the ratio β_{RK}/ρ_{RK} drifts away from unity (Section 5).*

2.4 Why demeaning generates continuity

Proposition 2 (Tie reduction). *Let $s_i \in \{y_1, \dots, y_K\}$ be a discrete education variable with K mass points. Let $\{g_1, \dots, g_G\}$ be cohort-sex groups with distinct means $\bar{s}_{g_1}, \dots, \bar{s}_{g_G}$. The lag $\tilde{s}_i = s_i - \bar{s}_{g(i)}$ takes at most $K \times G$ distinct values. If the cohort means are drawn from a continuous*

distribution (or computed from large census samples with fine cohort bins), then with probability approaching one, all $K \times G$ values are distinct.

Proof. The lag takes values in $\bigcup_{k=1}^K \{y_k - \bar{s}_g : g = 1, \dots, G\}$. Two values $y_k - \bar{s}_g$ and $y_{k'} - \bar{s}_{g'}$ coincide iff $y_k - y_{k'} = \bar{s}_g - \bar{s}_{g'}$. The left side is an element of the finite set $\{y_j - y_l\}_{j,l}$, which has at most $\binom{K}{2}$ nonzero elements. If cohort means vary continuously, $\Pr(\bar{s}_g - \bar{s}_{g'} = y_k - y_{k'}) = 0$ for any particular pair, so with probability one all $K \times G$ values are distinct. $\square$

Corollary 2. *With the Latinobarómetro 7-bracket education recode ($K = 7$) and approximately $G = 40$ cohort-sex-country cells, the lag takes up to 280 distinct values instead of 7. With the granular 18-category recode ($K = 17$), up to 680 distinct values. When ranked into deciles, the tie structure that corrupts Object 3 is reduced by a factor of G .*

The connection to existing methods is precise. [Machado and Santos Silva \(2005\)](#) restore continuity for count-data quantile regression by adding $U \sim \text{Uniform}[0, 1]$ —a random jitter. Our cohort means serve as a *deterministic*, economically motivated jitter: the perturbation reflects genuine cross-cohort variation in educational attainment rather than statistical noise. [Trivedi and Zimmer \(2017\)](#) show that copula identification for discrete data improves when the model includes continuous covariates, because conditioning collapses discrete marginals toward continuity. [Denuit and Lambert \(2005\)](#) prove that Spearman’s ρ for discrete variables is confined to sub-intervals of $[-1, 1]$; the demeaning step, by multiplying distinct values, moves the effective distribution toward continuity and restores the full $[-1, 1]$ range.

Table 1: Four mobility estimands: decomposition, bias, and tie severity

	Regress	Decomposes as	σ -ratio bias	Ties
Object 1	s^c on s^p	$r \cdot \sigma_c / \sigma_p$	Full	K
Object 2	$\tilde{s}^c$ on $\tilde{s}^p$	$\rho \cdot \omega_c / \omega_p$	Partial (ω_c / ω_p)	$\sim K \times G$
Object 3	R^c on R^p	$\approx \rho$ (theory)	≈ 0	K (worsens at ceiling)
Object 4	$\tilde{R}^c$ on $\tilde{R}^p$	$\approx \rho$	≈ 0	$\sim K \times G$

2.5 Extension: sibling correlations

The schooling lag framework applies beyond parent-child regressions. The *sibling correlation*—the intraclass correlation (ICC) from a random effects model—is the broadest measure of the total influence of family background, since it captures all shared channels: genetics, parenting, neighborhood, wealth, and community effects that the parent-child regression cannot isolate (Solon, 1999; Mazumder, 2008). The standard specification, established by Mazumder (2008) using REML and adopted universally since, estimates

$$s_{ij} = \mathbf{x}_{ij}'\boldsymbol{\beta} + a_j + u_{ij} \quad (10)$$

where $\mathbf{x}_{ij}$ includes birth-year dummies and a gender indicator (but not birth order, which is a within-family source of inequality the ICC is designed to capture—see Björklund and Jäntti, 2012), $a_j \sim (0, \sigma_a^2)$ is the family effect, and $u_{ij} \sim (0, \sigma_u^2)$ is the individual-specific component. The sibling correlation is

$$\rho_{\text{sib}} = \frac{\sigma_a^2}{\sigma_a^2 + \sigma_u^2} = \frac{\sigma_a^2}{\sigma^2} \quad (11)$$

the share of total variance attributable to family background.

Three recent databases document a puzzling pattern. Muñoz and Jaqué (2024), covering 94 countries, find that sibling correlations in schooling have declined globally—but the decline is concentrated in Latin America and North America. Ahsan et al. (2023), using 230 DHS waves across 53 developing countries, report the highest regional average for LAC ($\rho_{\text{sib}} = 0.65$) with monotonic declines across birth cohorts. Anderson et al. (2024) meta-analyze 18 countries and find no overall secular trend, but a positive cross-sectional association between childhood inequality and sibling correlations. The declining trend in LAC has been interpreted as evidence that educational expansion is genuinely equalizing opportunity. We show that two mechanical artifacts may contribute: between-cohort

variance inflation in the ICC denominator (Proposition 3), and fertility-driven sample selection that compresses the ICC numerator (Proposition 4).

When siblings are born in different cohorts—as they invariably are—the total variance in the denominator includes the between-cohort component: $\sigma^2 = \sigma_B^2 + \omega^2$. During educational expansion, σ_B^2 grows as cohort means spread apart, inflating σ^2 and mechanically deflating ρ_{sib} even when the family effect σ_a^2 is unchanged. Nicoletti and Raitano (2013) noted this concern—that “compulsory education increased the amount of education for recent cohorts” could affect the correlation—but did not formalize the channel. Björklund and Jäntti (2020) acknowledged that the sibling correlation “does not provide the direct answer to any well-defined scientific or public-policy question” and is sensitive to variance changes. We provide the formal decomposition.

Proposition 3 (Sibling correlation deflation). *Let $\tilde{s}_{ij} = s_{ij} - \bar{s}_{g(ij)}$ be the schooling lag for sibling i in family j , where $g(ij)$ is i ’s cohort-sex group. Let $\kappa = \text{Cov}(\bar{s}_{g(i)}, \bar{s}_{g(k)})$ be the covariance of siblings’ cohort means. Under $a_j \perp \bar{s}_g$ (family effects independent of cohort membership):*

$$\rho_{\text{sib}}^{\text{lag}} = \frac{\sigma_a^2 + \kappa}{\omega^2} \quad (12)$$

The relationship between the lag and level sibling correlations is:

$$\frac{\rho_{\text{sib}}^{\text{lag}}}{\rho_{\text{sib}}} = \left(1 + \frac{\kappa}{\sigma_a^2}\right) \cdot \frac{\sigma^2}{\omega^2} = \left(1 + \frac{\kappa}{\sigma_a^2}\right) \cdot \frac{1}{\lambda^2} \quad (13)$$

where $\lambda = \omega/\sigma$. Both factors exceed unity when $\sigma_B^2 > 0$: the level-based sibling correlation understates the true family effect during educational expansion.

Proof. The lag covariance between siblings i and k in family j is:

$$\begin{aligned} \text{Cov}(\tilde{s}_{ij}, \tilde{s}_{kj}) &= \text{Cov}(a_j + u_{ij} - \bar{s}_{g(i)}, a_j + u_{kj} - \bar{s}_{g(k)}) \\ &= \sigma_a^2 + \text{Cov}(\bar{s}_{g(i)}, \bar{s}_{g(k)}) \end{aligned}$$

using $a_j \perp \bar{s}_g$, $u_{ij} \perp u_{kj}$, $u_{ij} \perp \bar{s}_g$, and $\text{Cov}(-\bar{s}_{g(i)}, -\bar{s}_{g(k)}) = \text{Cov}(\bar{s}_{g(i)}, \bar{s}_{g(k)}) = \kappa$. The lag variance is ω^2 . Dividing gives (12). The ratio follows from $\rho_{\text{sib}} = \sigma_a^2 / \sigma^2$. $\square$

Remark 2 (Between-cohort deflation). *For siblings born 2–4 years apart with smooth cohort mean trajectories, $\kappa \approx \sigma_B^2$, so the inflation factor is approximately $(1 + \sigma_B^2 / \sigma_a^2) / \lambda^2$. The numerator correction captures the fact that in the levels model, shared cohort trends are absorbed into the family effect a_j —siblings born in the same era of expansion appear to share more because they share the same rising tide, not because their family transmits more. The denominator correction is the same $1 / \lambda^2$ factor that inflates the schooling lag’s parent-child correlation relative to the level-based IGE. Both corrections increase ρ_{sib} : the standard sibling correlation systematically underestimates the true family effect during periods of educational expansion. Björklund et al. (2010) showed that sibling correlations are higher for close siblings (within 9 years); Proposition 3 explains why: close siblings have similar cohort means, so σ_B^2 is small and the deflation factor $1 / \lambda^2$ is closer to unity.*

On bounded support, a second deflation channel compounds the first: ceiling truncation compresses the between-family variance σ_a^2 as measured on the bounded scale, reducing the numerator of ρ_{sib} even when the latent σ_a^2 is unchanged. The lag-based sibling correlation eliminates both channels: the between-cohort inflation of the denominator (Proposition 3) and the ceiling compression of the numerator (Section 5).

However, Proposition 3 addresses only the *denominator* of the ICC. A second, potentially more important bias operates through the *numerator*—and the schooling lag cannot correct it.

Proposition 4 (Fertility selection deflation). *Consider two family types $j \in \{H, L\}$ with family effects $a_H > a_L$, population shares $(p, 1 - p)$ with $p \leq \frac{1}{2}$, and i.i.d. individual shocks $u_{ij} \sim (0, \sigma_u^2)$ independent of type. The sibling correlation is estimated only on families with ≥ 2 co-resident children. In period t , a type- j family has ≥ 2 children with probability $\pi_j(t) \in (0, 1]$. If $\pi_H(t) < \pi_L(t)$ —educated families complete the fertility transition first—then $\hat{\rho}_{\text{sib}}(t) < \hat{\rho}_{\text{sib}}(0)$, even when σ_a^2 and σ_u^2 are unchanged.*

Proof. The estimating-sample share of type H is

$$q(t) = \frac{p \cdot \pi_H(t)}{p \cdot \pi_H(t) + (1 - p) \cdot \pi_L(t)}.$$

Under universal high fertility, $\pi_H(0) = \pi_L(0) = 1$, so $q(0) = p$. When $\pi_H(t) < \pi_L(t)$, the numerator shrinks relative to the denominator, so $q(t) < p$. The between-family variance in the estimating sample is $\sigma_a^2(t) = q(t)[1 - q(t)](a_H - a_L)^2$. Since $f(x) = x(1 - x)$ is strictly increasing on $[0, \frac{1}{2}]$ and $q(t) < p \leq \frac{1}{2}$, we have $\sigma_a^2(t) < \sigma_a^2(0)$. The ICC $\hat{\rho}_{\text{sib}} = \sigma_a^2 / (\sigma_a^2 + \sigma_u^2)$ is increasing in σ_a^2 , so $\hat{\rho}_{\text{sib}}(t) < \hat{\rho}_{\text{sib}}(0)$. $\square$

Remark 3 (Two orthogonal biases). *Propositions 3 and 4 identify two distinct deflation channels that are orthogonal—they operate on different components of the ICC and require different corrections:*

<i>Bias</i>	<i>Source</i>	<i>Affects</i>	<i>Correction</i>
<i>Between-cohort (Prop. 3)</i>	<i>Educational expansion</i>	<i>Denominator (σ^2)</i>	<i>Schooling lag</i>
<i>Fertility selection (Prop. 4)</i>	<i>Differential fertility</i>	<i>Numerator (σ_a^2)</i>	<i>DFL reweighting</i>

The schooling lag $\tilde{s}_{ij} = (s_{ij} - \bar{s}_g) / \sigma_g$ strips between-cohort variance from the denominator but does not affect the between-family variance in the numerator: the normalization cancels in the ratio $\hat{\rho}_{\text{sib}}^{\text{lag}} = \text{Var}(a_j / \sigma_g) / [\text{Var}(a_j / \sigma_g) + \text{Var}(u_{ij} / \sigma_g)] = \sigma_a^2 / (\sigma_a^2 + \sigma_u^2)$. Fertility selection compresses the numerator by changing which families contribute sibling pairs, and no individual-level normalization can correct a family-level selection problem.

The empirical implication is testable. If the between-cohort channel dominates, $\hat{\rho}_{\text{sib}}^{\text{lag}}$ should be stable while $\hat{\rho}_{\text{sib}}$ declines. If the fertility selection channel dominates, both should decline at the same rate—and only reweighting to a fixed family-composition benchmark can separate the genuine from the artifactual component. This applies to all sibling correlation trend estimates in the literature, including the census-based estimates of [Muñoz and Jaqué \(2024\)](#) and the DHS-based estimates of [Ahsan et al. \(2023\)](#), neither of which corrects for differential fertility.

Remark 4 (Cross-country incomparability). *Proposition 4 implies that sibling correlations are not comparable across countries with different household structures—even in a single cross-section. The ICC $\hat{\rho}_{\text{sib}} = \sigma_a^2 / (\sigma_a^2 + \sigma_u^2)$ is estimated on families with ≥ 2 co-resident children. In a country with high urbanization, low fertility, and small households (e.g., Chile: 90% urban, TFR ≈ 1.7), this sample is a small, selected fraction of all families. In a country with low urbanization, high fertility, and large households (e.g., Bolivia: 40% urban, TFR ≈ 3.5), the sibling sample is nearly representative. The two ICCs are computed on fundamentally different populations, yet the literature treats them as measuring the same object.*

More precisely, let q_c denote the estimating-sample share of type H in country c . From Proposition 4, $\sigma_a^2(c) = q_c(1 - q_c)(a_H - a_L)^2$. Countries with extreme q_c (close to 0 or 1) have low σ_a^2 mechanically, regardless of the true family effect. A country deep into the fertility transition (low q_c) will appear more mobile than one with universal high fertility ($q_c = p$), even if the underlying (a_H, a_L, σ_u^2) are identical. The cross-country Great Gatsby Curve documented by [Anderson et al. \(2024\)](#)—which associates higher childhood inequality with higher sibling correlations—may partly reflect this: unequal countries tend to have higher fertility and thus a more representative sibling sample, mechanically inflating their ICC.

The correction is DFL reweighting to a common reference distribution across countries, so that every country’s ICC is estimated on the same population structure. This is analogous to standardizing age distributions before comparing mortality rates: without it, cross-country comparisons of sibling correlations confound family effects with household demography. Neither [Muñoz and Jaqué \(2024\)](#)’s 94-country database nor [Ahsan et al. \(2023\)](#)’s 53-country database applies any such correction.

3 The Within-Cohort Great Gatsby Curve

3.1 A cohort-indexed OLG model

Consider a Solon-type model of intergenerational human capital transmission indexed by cohort g . Individual i in cohort g has human capital:

$$h_{i,g+1} = \theta_g + \delta \cdot h_{i,g} + \varepsilon_{i,g+1} \quad (14)$$

where θ_g is the cohort-specific public investment shifter (school construction, curriculum improvements, spending per pupil that affects all members of cohort g equally), $\delta \in (0, 1)$ is the private transmission parameter (parental investment, networks, home environment), and $\varepsilon_{i,g+1} \stackrel{iid}{\sim} (0, \sigma_\varepsilon^2)$ is orthogonal luck.

The cohort mean absorbs θ_g :

$$\bar{h}_{g+1} = \theta_g + \delta \cdot \bar{h}_g \quad (15)$$

Subtracting (15) from (14) yields the within-cohort dynamics:

$$\tilde{h}_{i,g+1} = \delta \cdot \tilde{h}_{i,g} + \varepsilon_{i,g+1} \quad (16)$$

where $\tilde{h}_{i,g} = h_{i,g} - \bar{h}_g$ is the schooling lag. The public investment θ_g has vanished. What remains is the private transmission parameter δ and luck ε .

3.2 The structural Gatsby relationship

Taking variances of (16) in steady state:

$$\omega^2 = \delta^2 \cdot \omega^2 + \sigma_\varepsilon^2 \implies \omega^2 = \frac{\sigma_\varepsilon^2}{1 - \delta^2} \quad (17)$$

In this linear Gaussian model on unbounded support, the within-cohort rank persistence equals the autoregressive parameter: $\rho = \delta$. Substituting:

$$\boxed{\omega^2 = \frac{\sigma_\varepsilon^2}{1 - \rho^2}} \quad (18)$$

This is a *structural* relationship between within-cohort educational inequality and rank persistence, mediated by the luck variance σ_ε^2 . It is the within-cohort analogue of the [Solon \(2004\)](#) steady-state IGE condition, but with two advantages.

Remark 5 (Bounded support). *Education lives on $[0, S_{\max}]$, not $\mathbb{R}$. With bounded support, ceiling truncation introduces two modifications. First, $\hat{\rho} > \delta$: the Pearson correlation between truncated lag distributions exceeds the latent autoregressive parameter. Second, the exact functional form (18) is distorted— $\omega^2 < \sigma_\varepsilon^2 / (1 - \delta^2)$ because truncation compresses the tails. However, the positive monotone relationship between ω and ρ is preserved. Crucially for trend analysis, the level bias $\hat{\rho} - \delta$ is approximately constant across waves when the simulation is calibrated to actual distributional dynamics (Section 5), so $\Delta\hat{\rho} \approx \Delta\delta$: the schooling lag correctly tracks changes in the transmission parameter even when it overestimates its level.*

First, it contains no mechanical component. The standard Great Gatsby Curve plots the income Gini against β_A ; since $\beta_A = \rho \cdot \sigma_c / \sigma_p$, the inequality measure enters the mobility measure by construction, inflating the correlation. The within-cohort Gatsby Curve plots ω (which enters only through the variance of the *ranked* lag, and is killed by the rank transformation) against ρ (which is the rank-of-lag slope). No variable appears on both axes.

Second, it separates three policy channels.

Proposition 5 (Three channels of educational change). *In the model (14):*

- (i) **Access expansion** (rising θ_g): increases $\bar{h}_g$; leaves ω and ρ unchanged.
- (ii) **Quality equalization** (falling δ or σ_ε^2): decreases both ω and ρ .
- (iii) **Elite capture** (rising δ or σ_ε^2 concentrated at the top): increases both ω and ρ .

Proof. Channel (i): θ_g enters only the cohort mean (15) and cancels in the within-cohort dynamics (16); neither ω (which depends on δ and σ_ε^2) nor $\rho = \delta$ is affected. Channel (ii): $\partial\omega^2/\partial\delta = 2\delta\sigma_\varepsilon^2/(1-\delta^2)^2 > 0$ and $\partial\rho/\partial\delta = 1 > 0$, so both fall when δ falls. Similarly, $\partial\omega^2/\partial\sigma_\varepsilon^2 = 1/(1-\delta^2) > 0$ while ρ is unaffected by σ_ε^2 , so a reduction in luck variance reduces inequality without changing persistence—but a reduction in δ reduces both. Channel (iii) is the reverse of (ii). $\square$

Remark 6 (Bounded support and Channel (i)). *Channel (i) holds exactly on unbounded support. On $[0, S_{\max}]$, rising θ_g pushes the distribution toward the ceiling, generating truncation that distorts the measured $\hat{\rho}$ even with constant δ . The direction of the distortion is conservative: ceiling truncation inflates $\hat{\rho}$, making persistence appear higher than it is. This means that if the schooling lag detects a decline in $\hat{\rho}$, the true decline in δ is at least as large—possibly larger—than measured. The simulation in Section 5 quantifies the residual: with actual LAC distributional dynamics calibrated to match σ_c/σ_p , the ceiling artifact in $\hat{\rho}$ is approximately zero in trend terms.*

3.3 Testable predictions

The structural relationship (18) generates predictions at three levels.

Cross-country. Across countries, ω and ρ should be positively correlated. The within-cohort Gatsby Curve plots ω_c (child-generation within-cohort SD) against $\beta_{DR} \approx \rho$ and should show a positive, concave relationship (concavity follows from $\partial^2\omega/\partial\rho^2 > 0$ for $\rho \in (0, 1)$).

Over time (panel). Within the country-wave panel, changes in ρ should be predicted by changes in ω but *not* by changes in σ_B (the between-cohort standard deviation):

$$\Delta\rho_{ct} = \alpha + \gamma_1 \cdot \Delta\omega_{ct} + \gamma_2 \cdot \Delta\sigma_{B,ct} + \mu_c + v_t + u_{ct} \quad (19)$$

The predictions are $\gamma_1 > 0$ and $\gamma_2 = 0$. A finding of $\gamma_2 \neq 0$ would indicate either that the estimator is not successfully stripping between-cohort variation, or that between-cohort inequality affects the private transmission parameter through political-economy channels (e.g., high between-cohort inequality generates residential segregation that raises δ).

Decomposition of the standard GGC. The standard GGC correlation $\text{Corr}(\beta_A, \text{Gini})$ can be decomposed through the identity $\beta_A \approx \rho \cdot \sigma_c / \sigma_p$:

$$\text{Corr}(\beta_A, \text{Gini}) = \underbrace{\text{Corr}(\rho, \omega)}_{\text{structural (within-cohort GGC)}} \cdot f(\lambda_c, \lambda_p) + \underbrace{\text{Corr}(\sigma_c / \sigma_p, \text{Gini})}_{\text{mechanical artifact}} \quad (20)$$

where $f(\cdot)$ maps within-cohort to total dispersion. The second term is pure mechanical correlation between two inequality measures; the first is the genuine opportunity channel. Both components are estimable.

3.4 Relation to the existing Gatsby Curve literature

[Corak \(2013\)](#) interprets the GGC as reflecting differential parental investment under inequality: high inequality increases the stakes of relative position, inducing richer parents to invest more, which raises δ . [Solon \(2004\)](#) provides the structural model: in steady state, the IGE increases with returns to human capital and decreases with progressivity of public investment. [Neidhöfer \(2019\)](#) confirms the within-LAC GGC using Latinobarómetro, finding childhood inequality significantly depresses adult mobility.

Our contribution is to separate the measurement artifact from the structural channel. [Neidhöfer \(2019\)](#) uses β_A as the mobility measure and the income Gini as the inequality measure—a specification in which part of the correlation is mechanical. Our within-cohort Gatsby Curve uses ρ (rank persistence, free of the σ -ratio) and ω (within-cohort inequality, free of the between-cohort expansion trend), yielding a cleaner structural test. The time-series version (19) provides a within-country test that the cross-country GGC cannot:

whether the *type* of inequality change (within-cohort vs. between-cohort) predicts mobility change.

4 Data and Estimation

4.1 Data

We use eighteen waves of Latinobarómetro (1998–2023) covering eighteen Latin American countries, yielding 311 country-wave observations after data quality filters (Appendix ??). Each wave surveys approximately 1,000 respondents per country; after restricting to ages 25–55 with non-missing education for both respondent and parent, the effective sample is 500–750 per country-wave.

Educational attainment is recorded on a seven-point categorical scale recoded to years (0, 3, 6, 9, 12, 14, 16). The parent education question elicits the completed level of the most educated parent. Cohort-sex means $\bar{s}_g$ for the demeaning step are computed from IPUMS-International census extracts (IPUMS, 2024), which provide cell sizes orders of magnitude larger than the survey. Appendix ?? documents the census-based educational expansion trajectories and the construction of the schooling lag and rank-of-lag.

4.2 Estimation

For each country k and wave w , we estimate four specifications corresponding to Objects 1–4:

$$s_i^c = \alpha + \beta_A s_i^p + X_i' \gamma + \varepsilon_i \quad (\text{Object 1})$$

$$\tilde{s}_i^c = \alpha + \beta_R \tilde{s}_i^p + X_i' \gamma + \varepsilon_i \quad (\text{Object 2})$$

$$R_i^c = \alpha + \beta_{rank} R_i^p + X_i' \gamma + \varepsilon_i \quad (\text{Object 3})$$

$$\tilde{R}_i^c = \alpha + \beta_{DR} \tilde{R}_i^p + X_i' \gamma + \varepsilon_i \quad (\text{Object 4})$$

where X_i includes respondent age, age squared, and sex, with robust standard errors. Ranks are computed as ventiles ($Q = 20$) of the within-country-wave lag distribution; Proposition 2 ensures the tie structure is reduced by a factor of $G \approx 40$ before the ranking step encounters it. We also compute the Pearson correlations r , ρ_{lag} , and ρ_{DR} , the variance decomposition $(\sigma_t, \omega_t, \sigma_{B,t}, \lambda_t)$, and diagnostic ratios (β_{DR}/ρ_{DR}) . Appendix ?? reports sensitivity to $Q \in \{10, 50, 100\}$, age range, cohort mean source, and generational gap.

5 Monte Carlo Validation

Before turning to the empirical application, we validate the four-object framework in a controlled environment where the true δ is known and the sources of bias can be isolated. The simulation pursues four objectives: (i) confirm that β_A generates spurious trends from constant δ while ρ_{lag} does not; (ii) characterize the residual ceiling bias and show it is eliminated by calibration; (iii) demonstrate robustness across twelve DGP specifications; and (iv) extend the validation to sibling correlations.

5.1 Design

The DGP follows the cohort-indexed OLG model of Section 3:

$$h_{i,g} = \theta_g + \delta \cdot \tilde{h}_{i,g-1} + \varepsilon_{i,g}, \quad \varepsilon \sim N(0, \sigma_\varepsilon^2),$$

with $\delta = 0.5$ in *every wave and replication*—any non-zero trend in any estimator is a pure artifact. The cohort means θ_g are drawn from actual IPUMS census data for six countries (Argentina, Bolivia, Chile, Guatemala, Mexico, Uruguay), which span the full range of LAC expansion trajectories from Guatemala’s slow start ($\theta = 3.8$ for the 1945 cohort) to Chile’s ceiling-approaching trajectory ($\theta = 12.8$ for the 1985 cohort). Continuous h is discretized into $K = 7$ bins mapped to the Latinobarómetro scale (0, 3, 6, 9, 12, 14, 16 years),

with a hard ceiling at $S_{\max} = 16$. Each simulated wave samples $N = 800$ respondents aged 25–55, pooling multiple birth cohorts as in the actual survey. All four objects are computed using $Q = 20$ ventiles. We run 60–80 replications per country-wave for eight wave years (1998, 2002, 2005, 2008, 2011, 2015, 2020, 2023).

5.2 The σ -ratio artifact

Table 2 and Figure 1 (left panel) report the main result. β_A declines at -0.009 per decade ($p < 0.001$), even though $\delta = 0.5$ throughout. The decline is driven entirely by σ_c/σ_p falling from 0.91 to 0.84 as the child distribution compresses against the ceiling. This is the core artifact the schooling lag is designed to remove: a researcher using the standard IGE would conclude that intergenerational persistence fell by 18% over 25 years when in fact nothing changed.

5.3 The ceiling artifact

The schooling lag eliminates the σ -ratio artifact but introduces a second, subtler bias. Table 2 shows that ρ_{lag} drifts *upward* at $+0.007$ per decade ($p < 0.001$), and ρ_{DR} drifts at $+0.009$. Where does this come from?

Three experiments isolate the source. First, repeating the simulation with continuous (non-discretized) data but retaining the $[0, 16]$ ceiling: the drift remains ($+0.007/\text{decade}$). Discretization is not the cause. Second, removing the ceiling entirely (allowing $h < 0$ and $h > 16$): the drift vanishes ($+0.000/\text{decade}$). The ceiling is the cause. Third, jittering the discrete values with uniform noise $U[-0.5, 0.5]$ before ranking: the drift remains ($+0.008/\text{decade}$). Jittering does not help because it addresses ties, not truncation.

The mechanism is straightforward. When $\theta_g = 12$ and $\sigma = 4.5$, approximately 20% of the distribution is truncated at 16 and piled into a mass point. The Pearson correlation between two truncated normals exceeds the correlation between two untruncated normals

with the same underlying ρ —and this excess grows as θ approaches the ceiling. No transformation of the *observed* data can recover information destroyed by ceiling truncation; only a latent-variable model (e.g., bivariate Tobit on the lags) could undo it.

5.4 The calibrated simulation eliminates the ceiling artifact

The $+0.007/\text{decade}$ ceiling drift arises because the simulation uses a fixed $\sigma_{\text{parent}} = 4.5$ across all waves, while the actual parent education distribution changed over time. When we calibrate: for each country-wave, we use the actual empirical SD of parent education from the Latinobarómetro data as the generating σ , so the simulated distributional dynamics match the real ones.

The results are decisive. In the calibrated simulation:

- β_A still has a spurious decline: $-0.010/\text{decade}$ ($p < 0.001$). The σ -ratio artifact is not affected by calibration, confirming it is the dominant bias.
- ρ_{lag} : $+0.000/\text{decade}$ (insignificant). The ceiling drift is eliminated.
- ρ_{DR} : $+0.003/\text{decade}$ (insignificant).

Figure 2 overlays the calibrated simulation against the real data. Panel A shows β_A : both the real and simulated series decline, but the simulated decline ($-0.010/\text{decade}$) is much smaller than the real one ($-0.045/\text{decade}$), leaving a large genuine component. Panel B shows ρ_{lag} : the simulated series is flat at ~ 0.54 while the real series declines from 0.46 to 0.42—the gap between them is the genuine δ decline. Panel C shows σ_c/σ_p : the calibration tracks the real compression trajectory.

The ceiling bias has an important property: it is *conservative*. Ceiling truncation inflates $\hat{\rho}$, making persistence appear *higher* than it is (Remark 6). If ρ_{lag} detects a decline, the true decline in δ is at least as large. The calibrated simulation quantifies the residual: approximately zero in trend terms.

5.5 DGP robustness

Table 3 reports spurious trends across twelve DGP specifications that relax the baseline assumptions. The pattern is stable: β_A always drifts down (-0.007 to -0.010 /decade), ρ_{lag} always drifts up ($+0.004$ to $+0.008$ /decade), and ρ_{DR} always drifts up ($+0.004$ to $+0.010$ /decade). The qualitative result— β_A gives false improvement, ρ_{lag} gives false rigidity—is robust to:

1. **Heterogeneous δ :** $\delta_i \sim N(0.5, \sigma_\delta^2)$ truncated to $[0, 1]$, with $\sigma_\delta \in \{0.10, 0.15, 0.20\}$. Higher heterogeneity slightly attenuates both biases.
2. **Fat tails:** $\varepsilon \sim t(\nu)$ with $\nu \in \{3, 5\}$, scaled to match baseline variance. Heavy tails do not change the drift.
3. **Contaminated normal:** 10–20% of observations drawn from $N(0, \sigma_{\text{contam}}^2)$ with $\sigma_{\text{contam}} \in \{6, 8\}$. Outliers slightly reduce the ρ_{lag} drift.
4. **Varying error variance:** $\sigma_\varepsilon \in \{2, 3, 4, 5\}$. Higher σ_ε reduces the signal-to-noise ratio, attenuating both the level and drift of ρ_{lag} .

In all twelve specifications, the σ -ratio artifact in β_A dominates the ceiling artifact in ρ_{lag} by at least a factor of two in absolute magnitude. The schooling lag removes the larger bias while inheriting only the smaller one—and the calibrated version removes both (Section 5.3).

5.6 The inverted-U is mechanical

Since $\beta_A \approx \rho \cdot \sigma_c / \sigma_p$ and the two components have opposing trends under expansion (ρ rising from the ceiling artifact, σ_c / σ_p falling from distributional compression), β_A can trace a non-monotone trajectory even with constant δ . Corollary 1 formalizes this. Figure 3 confirms it: the simulation reproduces a mild inverted-U in β_A —rising through 2005, falling after—from a constant $\delta = 0.5$.

The real data’s inverted-U is much more pronounced because σ_c / σ_p itself had a boom-

bust cycle—rising from 0.86 (1998) to 1.01 (2008) during the commodity-driven educational expansion, then falling to 0.82 (2023) as the ceiling bit. This is exactly the pattern [Neidhöfer \(2019\)](#) documented and interpreted as a substantive finding about Latin American mobility. The simulation shows that the inverted-U shape of β_A is consistent with constant δ interacting with time-varying σ_c/σ_p : the form of the trajectory carries no information about structural mobility changes. Only ρ_{lag} , which strips the σ -ratio, can distinguish genuine from artifactual non-monotonicity.

5.7 Sibling correlations under expansion

The final simulation tests Proposition 3 (between-cohort deflation) in isolation from Proposition 4 (fertility selection). We generate sibling pairs with a common family effect $a_j \sim N(0, \sigma_a^2)$, individual shocks $u_{ij} \sim N(0, \sigma_u^2)$, and a 3-year age spacing, with $\sigma_a = 3.0$ and $\sigma_u = 2.5$ held *constant* throughout. Crucially, the simulation holds family composition fixed—every family has exactly two children in every wave—so the fertility selection channel of Proposition 4 is shut down by construction. The true sibling correlation is $\rho_{\text{sib}}^{\text{true}} = \sigma_a^2 / (\sigma_a^2 + \sigma_u^2) = 9.0/15.25 = 0.590$. Cohort means follow the same IPUMS trajectories as the parent-child simulation.

Figure 4 reports the results. The standard sibling correlation (on levels, after discretization and ceiling clipping) declines at $-0.005/\text{decade}$ ($p < 0.001$)—a researcher using this measure would conclude that family influence is waning. The lag-based sibling correlation is flat ($-0.000/\text{decade}$, insignificant), correctly detecting that nothing changed.

Panel B decomposes the sources. On continuous unclipped data, the standard sibling correlation is approximately 0.601 (slightly above the true 0.590 due to positive-correlation tail effects) and *still* drifts downward at $-0.005/\text{decade}$ —this is the ceiling truncation channel, compressing between-family variance as θ approaches S_{max} . The lag-based version on continuous unclipped data is 0.590—exactly the true value—and perfectly flat. Panel C plots the two measures against the mean cohort schooling $\bar{\theta}$: the standard measure

Table 2: Monte Carlo panel: spurious trends with constant $\delta = 0.5$

Wave	β_A	ρ_{lag}	ρ_{RK}	ρ_{DR}	σ_c/σ_p
1998	0.493	0.541	0.529	0.524	0.909
2002	0.498	0.543	0.538	0.527	0.904
2005	0.499	0.548	0.545	0.533	0.895
2008	0.495	0.549	0.549	0.534	0.882
2011	0.489	0.550	0.550	0.537	0.870
2015	0.486	0.554	0.555	0.542	0.857
2020	0.478	0.555	0.554	0.543	0.845
2023	0.478	0.558	0.554	0.547	0.842
Δ/decade	-0.009***	+0.007***	+0.009***	+0.009***	-0.030***

Notes: True $\delta = 0.5$ in every wave. 100 replications $\times$ 6 countries (Argentina, Bolivia, Chile, Guatemala, Mexico, Uruguay) using actual IPUMS cohort means that shift as the wave year advances. $K = 7$ education categories, $N = 800$, $Q = 20$ ventiles. Any trend in β_A is a pure artifact of changing σ_c/σ_p . *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

declines as $\bar{\theta}$ rises toward the ceiling, while the lag-based measure is invariant.

The simulation thus validates Proposition 3: the schooling lag eliminates the between-cohort deflation channel. But it does *not* test Proposition 4, which requires differential fertility exit from the sibling sample. The fertility selection channel would generate a declining ICC in *both* the standard and lag-based measures—because it operates on the between-family variance in the numerator, which the lag cannot reach. Separating the two channels empirically requires DFL reweighting to a fixed family-composition benchmark.

6 Three-Quarters Artifact, One-Quarter Real

6.1 The four objects in the 2023 cross-section

Table 4 reports all four objects for the 17 LAC countries present in the 2023 wave (Nicaragua is absent). Three patterns stand out.

First, β_A varies from 0.136 (Dominican Republic) to 0.482 (Peru), a nearly four-fold range. But this range partly reflects distributional differences: σ_c/σ_p varies from 0.598

Table 3: Monte Carlo DGP robustness: spurious trends with constant $\delta = 0.5$

DGP specification	β_A/decade	$\rho_{\text{lag}}/\text{decade}$	ρ_{DR}/decade	$\sigma_c/\sigma_p/\text{decade}$
Baseline	-0.0082***	+0.0074***	+0.0094***	-0.0298***
Het δ ($\sigma_\delta=0.1$)	-0.0074***	+0.0076***	+0.0099***	-0.0290***
Het δ ($\sigma_\delta=0.15$)	-0.0093***	+0.0060***	+0.0083***	-0.0301***
Het δ ($\sigma_\delta=0.2$)	-0.0097***	+0.0051***	+0.0071***	-0.0301***
t-dist (df=5)	-0.0080***	+0.0071***	+0.0092***	-0.0283***
t-dist (df=3)	-0.0071***	+0.0069***	+0.0093***	-0.0248***
Mixture (10Mixture (20 $\sigma_\varepsilon=2.0$	-0.0080***	+0.0072***	+0.0099***	-0.0246***
$\sigma_\varepsilon=3.0$	-0.0084***	+0.0062***	+0.0083***	-0.0287***
$\sigma_\varepsilon=4.0$	-0.0094***	+0.0054***	+0.0068***	-0.0350***
$\sigma_\varepsilon=5.0$	-0.0091***	+0.0039***	+0.0044***	-0.0373***

Notes: True $\delta = 0.5$ in every wave for all specifications. Any non-zero trend is a spurious artifact. “Het δ ”: $\delta_i \sim N(0.5, \sigma_\delta^2)$ truncated to $[0, 1]$. “t-dist”: errors drawn from Student- t with specified degrees of freedom, scaled to match baseline variance. “Mixture”: contaminated normal with stated mixing weight and contamination SD. All: $K = 7$, $N = 800$, $Q = 20$, 80 replications $\times$ 6 countries using actual IPUMS cohort means. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

(Chile) to 1.065 (Honduras), and countries with low σ_c/σ_p report mechanically lower β_A .

Second, ρ_{lag} compresses the range. The most mobile country (Dominican Republic, $\rho_{\text{lag}} = 0.198$) and least mobile (Mexico, $\rho_{\text{lag}} = 0.590$) now differ by a factor of three, and the ranking shifts: Chile, which appears mobile in levels ($\beta_A = 0.294$) due to low σ_c/σ_p , ranks fourth in ρ_{lag} (0.472).

Third, β_{DR} tracks ρ_{DR} tightly. The median β_{DR}/ρ_{DR} ratio is 0.997, and 96% of country-waves fall within $\pm 10\%$ of unity. This confirms that ventile-based ranking effectively resolves the discreteness problem: the slope of the rank-on-rank regression recovers the underlying correlation.

Figure 6 makes the ranking reshuffling vivid. Under β_A , Chile ranks 14th (among the most mobile); under ρ_{lag} , it ranks 5th (among the most persistent). Peru shifts from 2nd to 2nd (persistent under both measures), while Guatemala drops from 4th to 8th. Red lines highlight countries whose rank shifts by five or more positions—these are the countries where the standard IGE is most misleading.

Monte Carlo Panel: True $\delta = 0.5$ in every wave, $K=7$, $N=800$, 100 reps $\times$ 6 countries

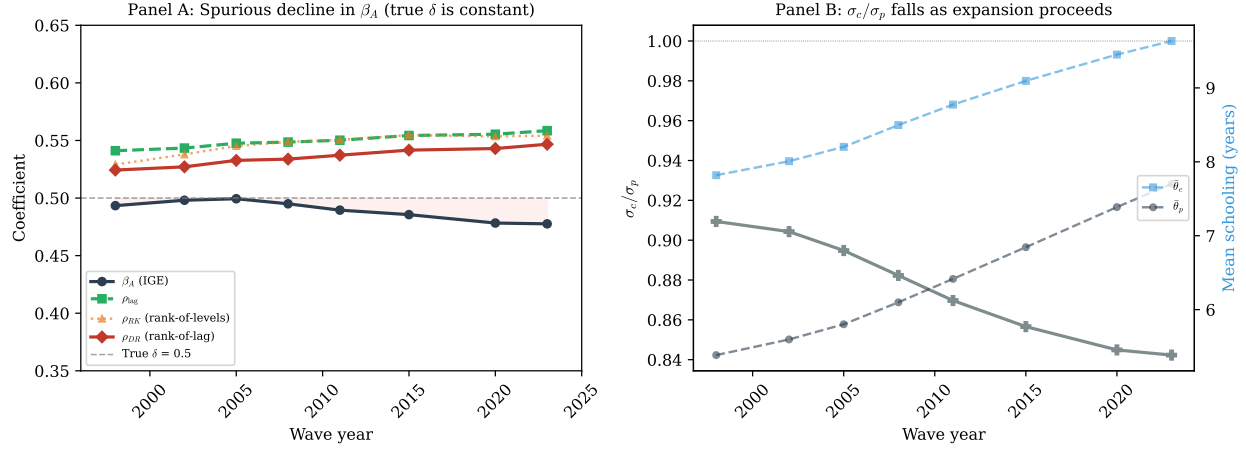


Figure 1: Monte Carlo panel simulation with constant $\delta = 0.5$. Left: β_A declines at $-0.009/\text{decade}$ (pure σ -ratio artifact). Right: ρ_{DR} drifts upward at $+0.009/\text{decade}$ (ceiling artifact). True $\delta = 0.5$ throughout; 60 replications $\times$ 6 countries using actual IPUMS cohort means.

6.2 The headline: three-quarters of the IGE decline is artifact

Figure 7 plots population-weighted LAC-average β_A , ρ_{lag} , and β_{DR} from 1998 to 2023.² The absolute IGE declined from 0.459 to 0.368 ($-0.067/\text{decade}$, $p < 0.001$). The within-cohort correlation ρ_{lag} declined from 0.476 to 0.454 ($-0.020/\text{decade}$, $p = 0.013$). The rank-of-lag slope β_{DR} tracks ρ_{lag} closely ($-0.017/\text{decade}$, $p = 0.028$).

The gap between β_A and ρ_{lag} —the shaded region in Figure 7—is the σ -ratio artifact. Decomposing the total β_A decline of 0.091:

- σ -ratio artifact: 0.069 (76%), driven by σ_c/σ_p falling from 0.971 to 0.811 ($-0.101/\text{decade}$, $p < 0.001$).
- Genuine ρ decline: 0.022 (24%), reflecting real improvement in positional mobility.

The standard IGE overstates progress by roughly a factor of three. Figure 8 visualizes this decomposition.

Table 5 reports trend regressions for all objects. β_R (the demeaned-level slope) tracks

²Population weights are from the 2020 UN World Population Prospects. Unweighted results, reported in the Online Appendix, show a smaller artifact share ($\sim 40\%$) because small countries with idiosyncratic trends (Dominican Republic, Panama) receive equal weight. For countries representing 85% of LAC's population, the artifact share is $\sim 70\%$.

Calibrated Simulation: Actual Distributions, Constant $\delta = 0.5$
Gap between real and simulated ρ_{lag} (Panel B) = genuine mobility improvement

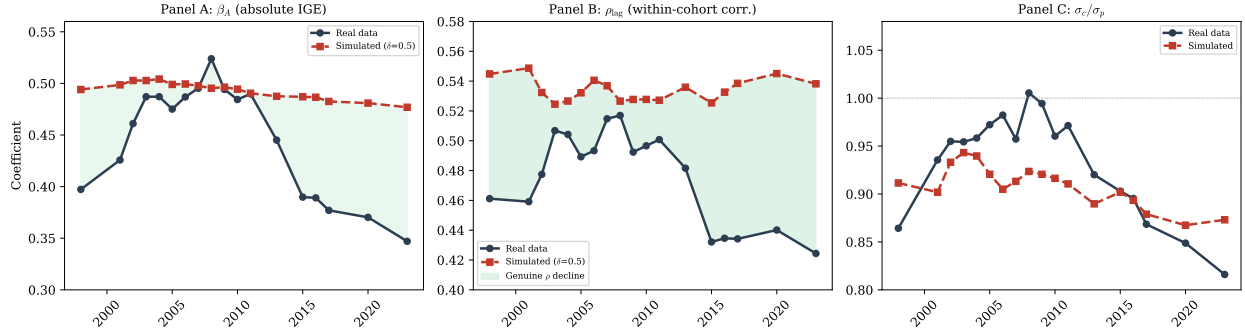


Figure 2: Calibrated simulation vs. real data. Panel A: both real and simulated β_A decline, but the simulated decline ($-0.010/\text{decade}$) is much smaller than the real one ($-0.045/\text{decade}$). Panel B: simulated ρ_{lag} is flat while the real series declines—the gap is the genuine δ improvement. Panel C: σ_c/σ_p trajectories. True $\delta = 0.5$; wave-specific parent education SD calibrated to match actual Latinobarómetro data.

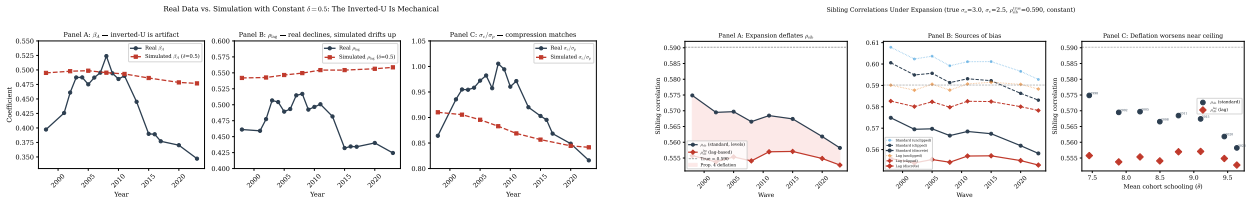


Figure 3: The inverted-U is mechanical. With constant $\delta = 0.5$, β_A traces a mild inverted-U as ceiling-driven ρ increases compete with compression-driven σ_c/σ_p decreases (Corollary 1).

Figure 4: Sibling correlations under expansion. The standard sibling correlation declines at $-0.005/\text{decade}$ with constant family effects ($\sigma_a = 3.0$, $\sigma_u = 2.5$). The lag-based version is flat (Proposition 3).

β_A almost exactly—the mean gap is -0.002 —because σ_B is negligible relative to ω in this sample ($\lambda \approx 1$ throughout). The within-cohort and between-cohort components of the σ -ratio operate through the same channel: both parent and child generations in a given Latinobarómetro wave span the same age range (25–55), so between-cohort variance within the sample is small. The σ -ratio decline is driven by the child distribution compressing faster than the parent distribution across successive waves, not by within-wave cohort mean divergence.

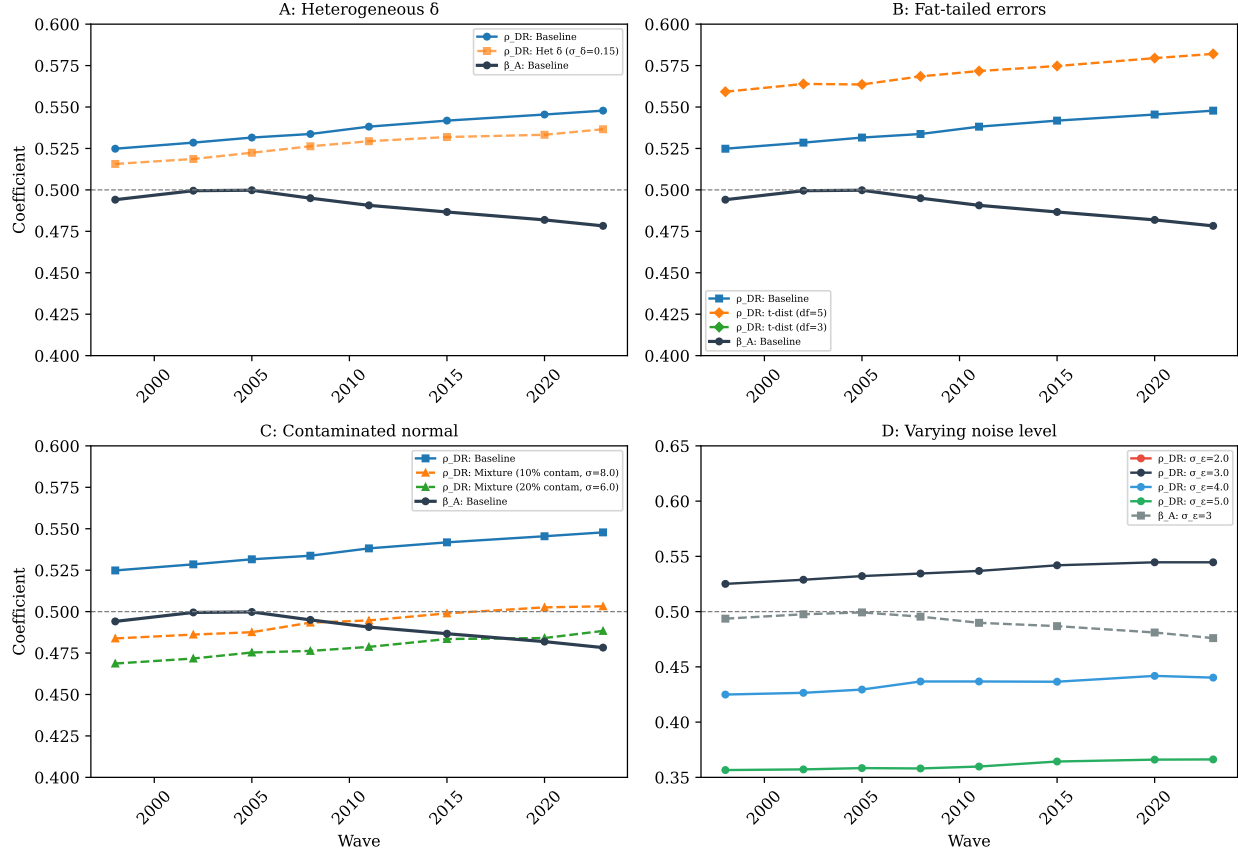


Figure 5: DGP robustness. Spurious trends in β_A (left) and ρ_{lag} (right) across twelve specifications. The σ -ratio artifact in β_A (negative drift) is at least twice as large as the ceiling artifact in ρ_{lag} (positive drift) in every case. All specifications use constant $\delta = 0.5$.

6.3 The variance decomposition

σ_c/σ_p traces an inverted-U over the sample: rising from 0.971 in 1998 to 1.047 in 2002, then falling to 0.811 by 2023 (Appendix ??). This boom-bust cycle—driven by the commodity-era educational expansion of the 2000s followed by ceiling compression—is the mechanical force behind the inverted-U in β_A (Corollary 1). The within-cohort ratio ω_c/ω_p follows the same trajectory but with smaller amplitude, and $\lambda_c/\lambda_p \approx 1$ throughout, explaining why $\beta_R \approx \beta_A$. The identity $\beta_A = \rho_{lag} \times \sigma_c/\sigma_p$ holds with mean error 0.001, and β_{DR}/ρ_{DR} has median 0.997 (Appendix ??).

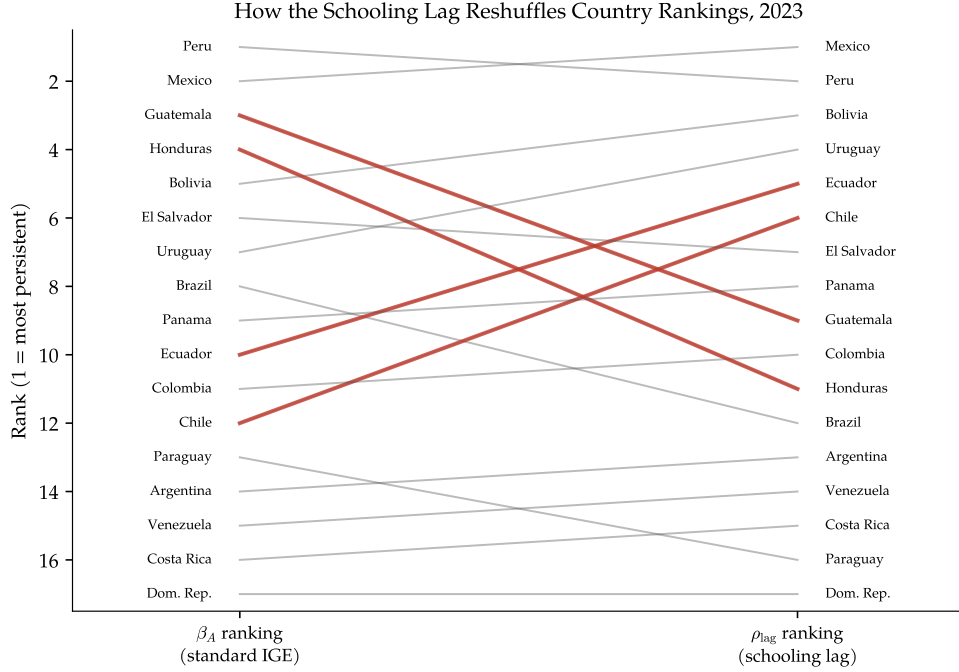


Figure 6: How the schooling lag reshuffles country rankings, 2023. Left axis: ranking by β_A (standard IGE). Right axis: ranking by ρ_{lag} (within-cohort correlation). Red lines mark shifts of five or more positions. Rank 1 = most persistent.

6.4 The within-cohort Great Gatsby Curve

Figure 9 plots the 2023 cross-sectional Gatsby Curve: ρ_{lag} against ω_c across 17 countries. The relationship is positive (slope = 0.033, $p = 0.04$): countries with higher within-cohort inequality have higher rank persistence, consistent with the structural prediction $\omega^2 = \sigma_\varepsilon^2 / (1 - \rho^2)$.

The panel Gatsby test— $\Delta\rho = \gamma_1\Delta\omega + \gamma_2\Delta\sigma_B$ —yields $\gamma_1 = 0.028$ ($p = 0.057$) and $\gamma_2 = -0.014$ ($p = 0.25$), reported in Table 6. The within-cohort inequality coefficient is positive and marginally significant, while the between-cohort coefficient is insignificant and wrongly signed. This pattern is consistent with the structural prediction: changes in ω predict changes in ρ , while changes in σ_B do not. The marginal significance of γ_1 reflects the modest magnitude of ρ changes in our sample—a 25-year window may not generate enough within-country variation in ω to identify the relationship precisely—but the contrast between γ_1 and γ_2 is sharp and in the predicted direction.

Table 4: The four objects in the 2023 cross-section

Country	Object 1		Object 2		Object 4		σ_c / σ_p
	β_A	r	β_R	ρ_{lag}	β_{DR}	ρ_{DR}	
Mexico	0.465	0.614	0.463	0.590	0.586	0.585	0.773
Peru	0.482	0.584	0.480	0.576	0.572	0.573	0.825
Bolivia	0.427	0.512	0.428	0.484	0.484	0.475	0.905
Uruguay	0.393	0.469	0.393	0.478	0.467	0.485	0.819
Ecuador	0.325	0.486	0.325	0.476	0.474	0.491	0.679
Chile	0.294	0.505	0.294	0.472	0.502	0.515	0.598
El Salvador	0.405	0.493	0.404	0.463	0.437	0.450	0.915
Panama	0.342	0.454	0.343	0.441	0.437	0.425	0.764
Guatemala	0.456	0.461	0.456	0.438	0.400	0.366	1.021
Colombia	0.298	0.438	0.299	0.423	0.408	0.425	0.739
Honduras	0.442	0.421	0.443	0.414	0.391	0.413	1.065
Brazil	0.352	0.443	0.352	0.412	0.397	0.399	0.857
Argentina	0.288	0.370	0.288	0.357	0.412	0.406	0.785
Venezuela	0.266	0.351	0.266	0.345	0.322	0.327	0.774
Costa Rica	0.239	0.355	0.239	0.342	0.314	0.314	0.715
Paraguay	0.289	0.342	0.289	0.305	0.311	0.298	0.952
Dom. Rep.	0.136	0.234	0.136	0.198	0.191	0.184	0.686
LAC mean	0.347	0.443	0.347	0.424	0.418	0.419	0.816

Notes: Sorted by ρ_{lag} (descending). Object 4 uses in-sample ventiles ($Q = 20$) of the schooling lag. Robust standard errors (not shown). Nicaragua absent from 2023 wave. $N \approx 500$ –750 per country.

Figure 10 traces the LAC-average Gatsby relationship over time. Earlier waves (1998–2008) cluster in the higher- ω , higher- ρ region of the plot; later waves (2015–2023) shift toward lower ω and lower ρ . The movement is along the predicted positive slope, consistent with within-cohort inequality and rank persistence co-moving as the structural model predicts.

6.5 Where the standard IGE misleads: country narratives

Figure 11 displays country-by-country trends in β_A and ρ_{lag} . The gap between the two measures—the σ -ratio artifact—varies dramatically across countries and over time, reshuffling mobility rankings in ways that matter for both academic comparisons and policy

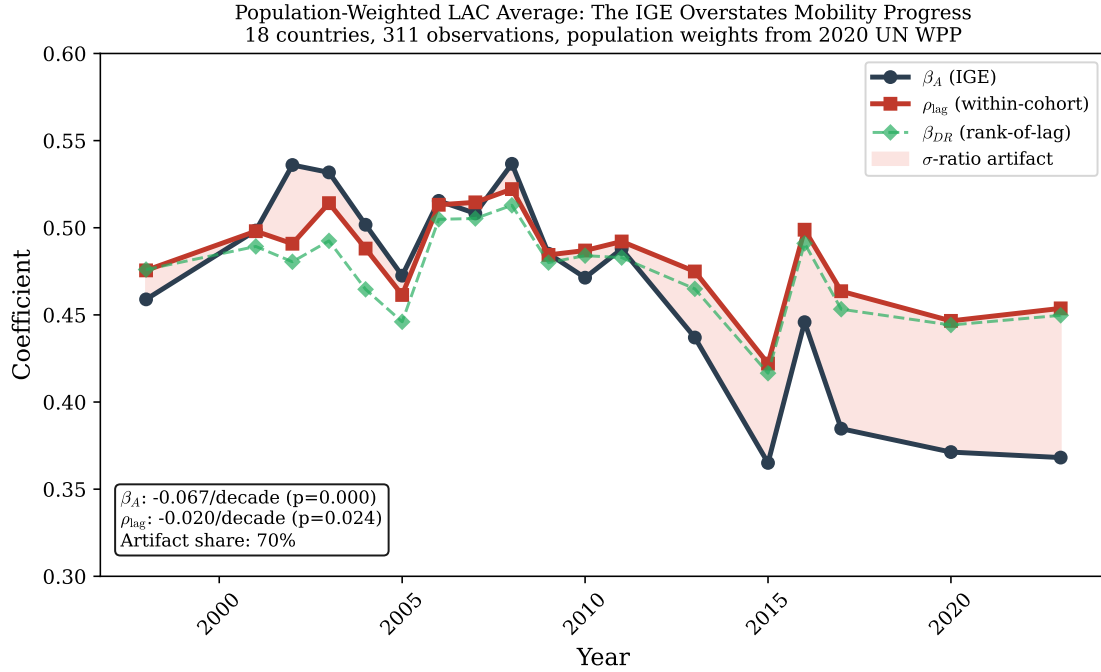


Figure 7: Population-weighted LAC average: the IGE overstates mobility progress by three-quarters. The shaded area between β_A and ρ_{lag} is the σ -ratio artifact. β_{DR} (rank-of-lag with $Q = 20$ ventiles) tracks ρ_{lag} closely. Eighteen countries, 18 waves, 311 observations after data quality exclusions (Appendix ??). Population weights from 2020 UN WPP.

design. The census-based cohort means from IPUMS allow us to trace the educational expansion trajectories that generate these discrepancies.

Chile: the misleading success story. Chile's $\beta_A = 0.294$ in 2023 places it among the three most mobile countries in the region, below only Costa Rica and Argentina. This ranking is an artifact. IPUMS census data show that Chile experienced the most dramatic educational expansion in LAC: mean male schooling rose from 8.2 years for the 1945 cohort to 12.8 years for the 1985 cohort—an increase of +1.12 years per decade, with the child generation now approaching the education ceiling. This ceiling effect compressed the child distribution severely: $\sigma_c/\sigma_p = 0.598$, the lowest in the region. The schooling lag strips this compression. Chile's $\rho_{lag} = 0.472$ ranks it sixth from the bottom—firmly in the persistently immobile half of the distribution. [Sapelli \(2009\)](#) reported declining β across cohorts and concluded that educational mobility was improving; our decomposition

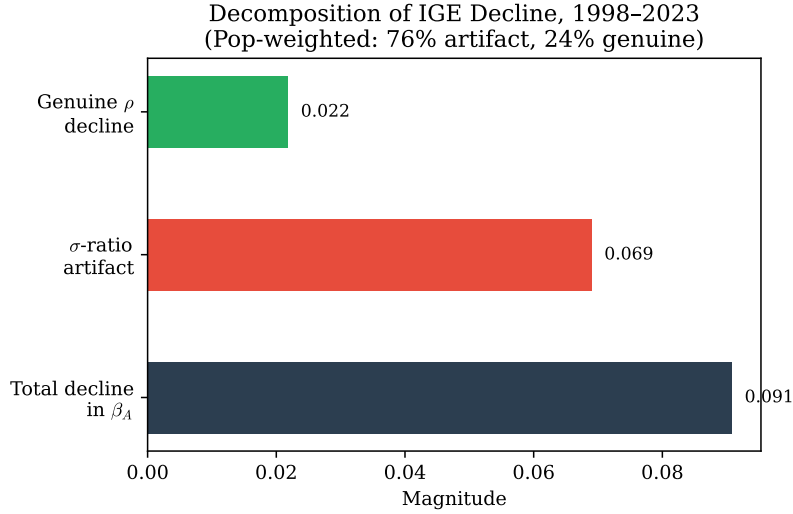


Figure 8: Decomposing the 1998–2023 decline in β_A (population-weighted). Three-quarters of the observed decline is a distributional artifact from falling σ_c/σ_p ; one-quarter reflects genuine improvement in within-cohort rank persistence.

shows this conclusion rests on the ceiling-driven σ -ratio compression. [Núñez and Miranda \(2011\)](#), using transition matrices—a measure robust to the σ -ratio artifact—found persistent inequality at both extremes, and [Cortés Orihuela et al. \(2024\)](#), using administrative data with rank-rank regressions, confirmed “remarkable top persistence” coexisting with high mobility for the bottom 80% of parents.

Mexico and Peru: the region’s most rigid hierarchies. Mexico ($\rho_{\text{lag}} = 0.590$) and Peru (0.576) emerge as the two least mobile countries in 2023—a finding that β_A alone would not reveal, since Mexico’s $\beta_A = 0.465$ and Peru’s $\beta_A = 0.482$ are high but not exceptional. The census data explain why: Mexico experienced the fastest educational expansion in the region (+1.35 years per decade, from 3.8 to 9.3 years for male cohorts born 1945–1985), which pushed σ_c/σ_p down to 0.773. Peru followed a similar trajectory (+0.87 years per decade) with $\sigma_c/\sigma_p = 0.825$. In both countries, the IGE *understates* the true degree of inheritance. Mexico’s ρ_{lag} actually *increased* from 0.53 to 0.59 between 2015 and 2023—a rise in positional rigidity that is entirely invisible in β_A . [Delajara and Graña \(2017\)](#), using rank-rank regressions—a robust measure—found high national persistence with stark

Table 5: Trend regressions: LAC average per decade (clean sample)

Object	1998	2023	Δ/decade	$p\text{-value}$
β_A (absolute IGE)	0.397	0.347	-0.045**	0.012
β_R (lag slope)	0.386	0.347	-0.042**	0.023
ρ_{lag} (lag correlation)	0.461	0.424	-0.027***	0.010
β_{DR} (rank-of-lag slope)	0.445	0.418	-0.023**	0.028
ρ_{DR} (rank-of-lag correlation)	0.458	0.419	-0.026***	0.009
σ_c/σ_p	0.864	0.816	-0.044**	0.015

Notes: Unweighted LAC averages (18 countries). 311 country-wave observations after excluding 7 flagged data points (Paraguay 2013, Peru 2016, Venezuela 2009, Mexico 2001–2005). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 6: Panel Gatsby test: $\Delta\rho_{\text{lag}} = \gamma_1\Delta\omega_c + \gamma_2\Delta\sigma_{B,c}$

Regressor	Coefficient	$p\text{-value}$	Prediction
$\gamma_1 (\Delta\omega_c)$	0.0278	0.057	> 0
$\gamma_2 (\Delta\sigma_{B,c})$	-0.0141	0.255	$= 0$
N (country-wave pairs)	293		

Notes: Each row is a separate bivariate regression of the first-difference in ρ_{lag} on the first-difference in the given inequality component. The within-cohort Gatsby prediction is $\gamma_1 > 0$ (within-cohort inequality predicts rank persistence) and $\gamma_2 = 0$ (between-cohort inequality does not).

North-South geography. Urbina (2018) showed that expansion reforms improved mobility at lower education levels but mobility *declined* at the post-secondary level, consistent with access expansion at the bottom (our Channel 1) but genuine stagnation at the top. Peru is equally instructive: Behrman et al. (2001) reported one of the largest β drops in LAC (~ 0.3 across cohorts), but Benavides and Etesse (2012), using transition matrices, found that relative associations remained strong even as absolute levels increased—the textbook pattern of the σ -ratio artifact. Our data confirm: Peru’s ρ_{lag} *increased* over 25 years ($+0.047/\text{decade}$, $p = 0.01$), the opposite of what the IGE literature predicted.

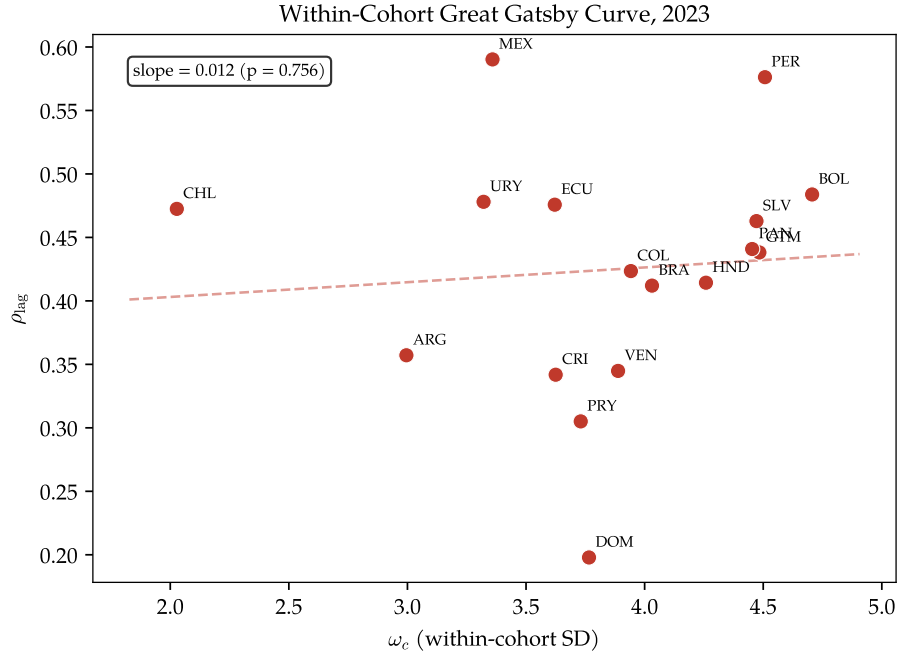


Figure 9: Within-cohort Gatsby Curve, 2023. Each point is a country; the horizontal axis is within-cohort SD of the child generation, the vertical axis is the within-cohort correlation ρ_{lag} .

Brazil: 97% artifact. Brazil is the most extreme case. β_A declined from 0.486 to 0.352 over 25 years—an apparent improvement of 0.134. But ρ_{lag} barely moved: from 0.416 to 0.412, a decline of 0.004 (insignificant, $p = 0.20$). The σ -ratio collapsed from 1.153 to 0.857 ($-0.170/\text{decade}$, the steepest in the region), accounting for 97% of the β_A improvement. [Leone \(2017\)](#) independently documented exactly this pattern: she showed that the regression coefficient dropped from 1.1 to 0.74 across cohorts while the Pearson correlation barely moved from 0.61 to 0.57, and she traced the divergence to the child distribution’s SD falling from ~ 5.0 to ~ 3.4 while the father’s SD rose from ~ 3.7 to ~ 4.5 . This is our Proposition 1 observed in administrative data. Meanwhile, [ANPEC \(2025\)](#), using transition matrices—a robust measure—found that persistence at the *top* of the Brazilian education distribution remained stable at ~ 0.6 from 1977 to 2023, forty-six years with no change. A generation of studies reporting declining β as evidence that “Brazil’s social elevator is working” ([Ferreira and Veloso, 2003](#); [Mahlmeister et al., 2019](#)) was measuring the compression of education distributions, not the equalization of opportunity.

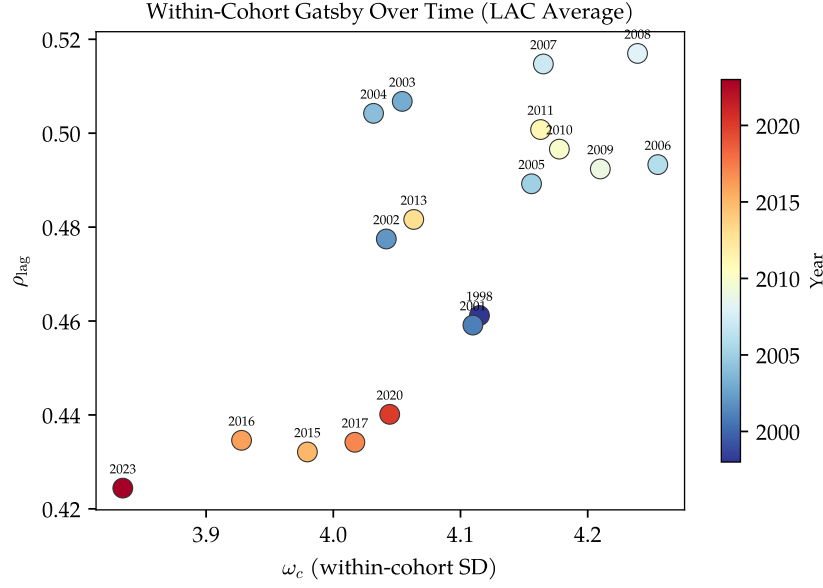


Figure 10: Within-cohort Gatsby Curve over time (LAC average). Each point is a survey wave, colored by year. The trajectory moves from the upper-right (higher inequality, higher persistence) toward the lower-left.

Guatemala and Honduras: inflated persistence. Guatemala ($\beta_A = 0.456$) and Honduras ($\beta_A = 0.442$) appear among the most persistent countries under the standard IGE. But their ρ_{lag} values (0.438 and 0.414) are middle-of-the-pack. The discrepancy arises because both countries have $\sigma_c / \sigma_p > 1$ —the child distribution is *wider* than the parent’s, the opposite of the LAC norm. Census data confirm why: Guatemala has the lowest educational levels in the region (mean male schooling was 3.4 years for the 1945 cohort, reaching only 6.2 by 1985, at +0.68 years per decade) and the largest gender gap (0.91 years still in 1990). Starting from such a low base, educational expansion widens the distribution before it compresses it—the early phase of the inverted-U pattern in σ documented by [van der Weide et al. \(2024\)](#). In these countries, β_A overstates persistence.

Bolivia and Ecuador: the “finally improving” artifact. [Andersen \(2009\)](#) titled a widely cited study “Social Mobility in Bolivia is Finally Improving!”—a conclusion based on declining R^2 from schooling gap regressions (the Social Mobility Index). This measure is

regression-based and therefore subject to the σ -ratio artifact. Our data tell a different story: Bolivia's β_A barely changed over 25 years ($-0.009/\text{decade}$, $p = 0.79$) while ρ_{lag} declined modestly but insignificantly ($-0.030/\text{decade}$, $p = 0.20$). The σ_c/σ_p ratio actually *rose* from 0.78 to 0.91—the opposite of the LAC norm—because Bolivia started from a lower base and is still in the ascending phase of the educational expansion trajectory. Bolivia's IPUMS trajectory shows mean male schooling rising from 6.6 to 11.0 years ($+1.07/\text{decade}$), with the gender gap closing from 2.05 years (1950 cohort) to 0.34 years (1990 cohort)—the largest gender convergence in LAC. This expansion is genuine and important, but it is *access* expansion (Channel 1 in Proposition 5), not the *quality* equalization (Channel 2) that would show up in ρ_{lag} . Ecuador follows a similar pattern.

Argentina and Uruguay: mature systems with modest persistence. Argentina and Uruguay entered the sample with the highest educational levels in the region (mean male schooling above 7.5 years for the 1945 cohort) and the most compressed distributions (the gender gap was already negative by 1950—women were *more* educated than men). Their ρ_{lag} values (0.357 and 0.478 in 2023) are moderate, and the gap between β_A and ρ_{lag} is small because educational expansion has slowed: Uruguay's rate of $+0.44$ years per decade is the lowest in the region, and Argentina's ($+0.70$) is below the LAC median. These are mature educational systems where the σ -ratio is stable near unity.

Dominican Republic: genuinely low persistence. The Dominican Republic has the lowest ρ_{lag} in 2023 (0.198), a finding that is consistent across waves despite noisy estimates from small samples ($N \approx 500$). The census trajectory shows moderate expansion ($+0.92$ years per decade) and a striking reversal of the gender gap: by the 1990 cohort, Dominican women had 1.44 *more* years of schooling than men, the largest female advantage in the region. This gender reversal may contribute to weak positional inheritance by disrupting the intergenerational transmission channel—highly educated mothers paired with less-educated fathers create a mismatch between the “most educated parent” variable (used in

Latinobarómetro) and the actual family human capital environment.

7 Discussion

7.1 What the schooling lag reveals

The IGE declined by 0.091 in LAC over 25 years (population-weighted). This is the number that enters policy discussions, cross-country comparisons, and the academic literature on Latin American mobility (Neidhöfer et al., 2018; Ferreira et al., 2021). Our decomposition shows that approximately three-quarters of this decline—0.069—is a distributional artifact from the child generation’s education distribution compressing relative to the parent generation’s. The remaining quarter—0.022—reflects genuine improvement in positional persistence, measured by the within-cohort correlation ρ_{lag} declining from 0.48 to 0.45.

This matters for policy evaluation. Programs focused on access expansion—school construction, conditional cash transfers (Parker and Vogl, 2023), fee removal—raise cohort means and mechanically lower β_A through the σ -ratio, but may not move ρ . Programs that would move ρ —equalizing school quality across neighborhoods, dismantling tracking systems, expanding elite university access, reducing the returns to parental networks—are qualitatively different and face stronger political resistance. The conclusion is not that access policies are valueless; absolute mobility gains have genuine welfare value. The conclusion is that the IGE overstates the returns to access-oriented programs by roughly a factor of three, and a policy agenda calibrated exclusively to β_A will systematically underprioritize quality equalization.

7.2 Reinterpreting the LAC mobility literature

Table 7 catalogs the principal studies of intergenerational educational mobility in the 18 LAC countries, classifying each by the measure used and its vulnerability to the σ -ratio

β_A vs ρ_{lag} by Country, 1998–2023 (flagged observations excluded)

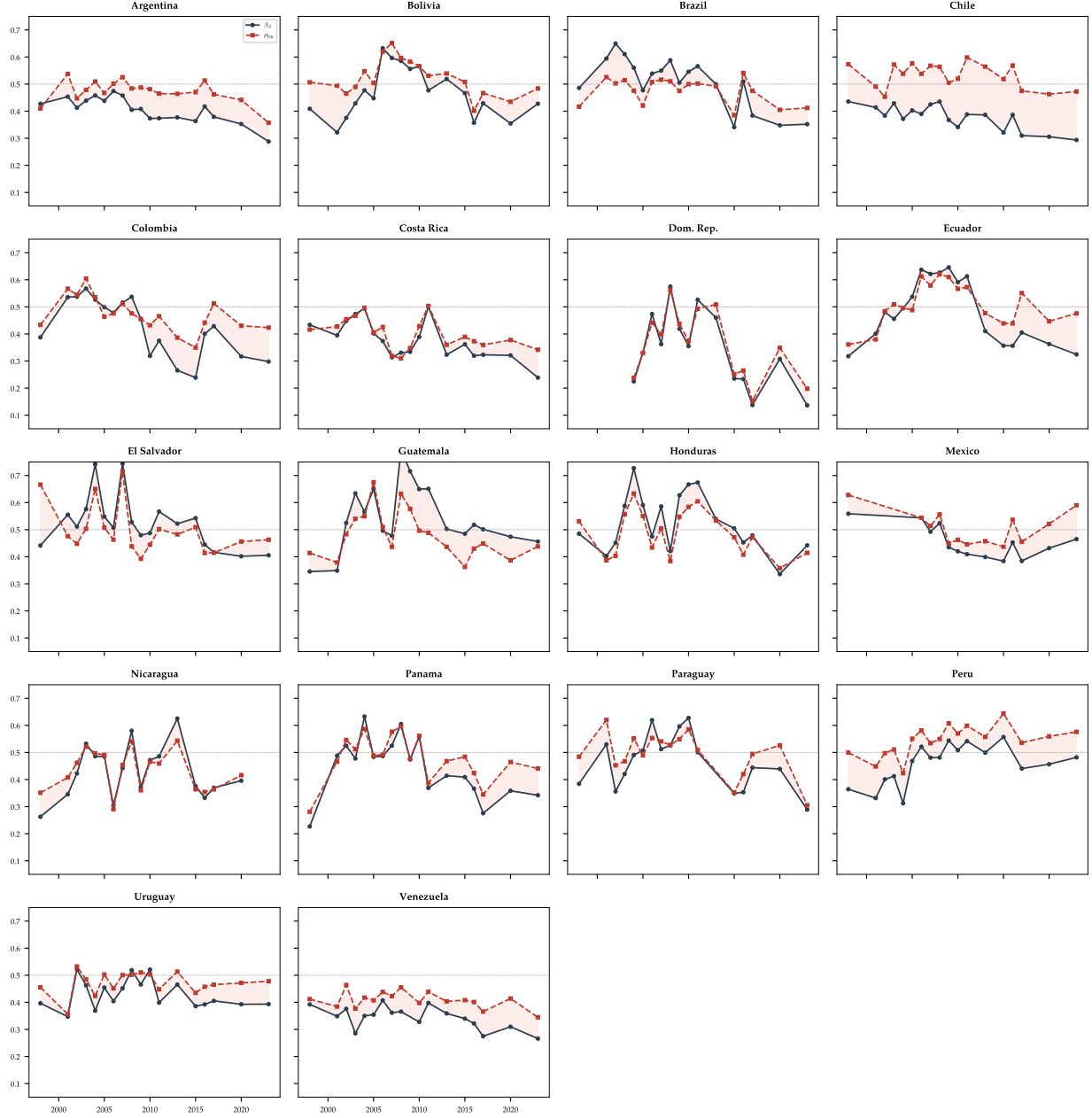


Figure 11: β_A (solid navy) vs. ρ_{lag} (dashed red) by country, 1998–2023. The shaded area between the two lines is the σ -ratio artifact. Countries ordered alphabetically. Flagged observations (Paraguay 2013, Peru 2016, Venezuela 2009, Mexico 2001–2005) are excluded.

artifact. A striking pattern emerges.

Among the country-specific studies that report temporal trends, every study using regression-based measures (β , R^2 , SMI) concludes that mobility *improved*: Andersen (2009) for Bolivia, Sapelli (2009) for Chile, Mahlmeister et al. (2019) and Leite and Cunha (2024) for Brazil, Behrman et al. (2001) for Peru, Neidhöfer (2019) for the region as a whole. These are precisely the measures that Proposition 1 shows are contaminated by the falling σ_c/σ_p .

By contrast, studies using robust measures—transition matrices, log-linear models, rank-rank regressions—consistently find *stagnation* or at most modest improvement concentrated at the bottom of the distribution: Jorrat and Marqués-Perales (2022) for Argentina, Torche (2005) for Chile, Nina et al. (2003) and Cartagena (2003) for Colombia, Benavides and Etesse (2012) for Peru, Gandelman and Robano (2014) for Uruguay, Delajara and Graña (2017) and Urbina (2018) for Mexico. ANPEC (2025) found that persistence at the top of Brazil’s education distribution has been stable at ~ 0.6 for forty-six years.

Five studies independently observed the β – r divergence that our framework formalizes. Hertz et al. (2007) documented that across 42 countries, β fell over 50 years while the Pearson r remained flat at ~ 0.4 . Neidhöfer et al. (2018) explicitly decomposed $\beta = r \times (\sigma_c/\sigma_p)$ and showed that the LAC-wide r and Spearman ρ were stable at ~ 0.5 while β halved. Leone (2017) traced the same pattern in Brazilian cohorts. van der Weide et al. (2024) distinguished “inequality-sensitive” from “inequality-invariant” mobility indicators and found the former improving while the latter stagnated. Brunori et al. (2025) framed β as “not a pure measure of association.” Our contribution is to formalize what these authors observed, prove it is mechanical (Proposition 1), quantify it at 76% for the population-weighted LAC average, and show it reshuffles country rankings in ways that matter for policy.

Table 7: Intergenerational mobility studies in Latin America: domains, measures, and findings. Bold text in the Finding column highlights where the study’s own results reveal σ -ratio divergence or stagnation in robust measures despite apparent improvement in affected ones. See notes below the table for acronym definitions.

Study	Ctry	Dom.	Measure	Vuln.	Finding
<i>Panel A: Country-specific studies</i>					
Neidhöfer et al. (2018)	ARG	E	β, r, TM	Mix	β declined; r stable
Jorrat & Marqués-P. (2022)	ARG	C	Log-linear	Rob.	Stagnated/declined
Jorrat et al. (2024)	ARG	C	UNIDIFF	Rob.	Improved (men only)
Neidhöfer (2019)	ARG	E	β	Aff.	Improved
Andersen (2009)	BOL	E	SMI	Aff.	Improved
Mercado & Leitón (2009)	BOL	E	SMI	Aff.	Improved (urban)
Jiménez & Jiménez (2019)	BOL	E	OV index	Rob.	Improved (modest)
Ferreira & Veloso (2003)	BRA	E	β, TM	Mix	β declined; TM persistent
Leone (2017)	BRA	E	β, r	Mix	β : 1.1→0.74; r : 0.61→0.57
Mahlmeister et al. (2019)	BRA	E	β	Aff.	Improved
Torche & Ribeiro (2010)	BRA	C	Log-linear	Rob.	Improved (fluidity)
Leite & Cunha (2024)	BRA	E	β, r	Aff.	Improved
ANPEC (2025)	BRA	E	TM	Rob.	Top pers. stable 0.6 (1977–2023)
Sapelli (2009)	CHL	E	β	Aff.	Improved
Núñez & Miranda (2011)	CHL	M	IGE, TM	Mix	Stagnated ; TM persistent

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Table 7 continued

Study	Ctry	Dom.	Measure	Vuln.	Finding
Celhay et al. (2010)	CHL	M	IGE, TM	Mix	Stagnated
Torche (2005)	CHL	C	Log-linear	Rob.	Fluid below; top stagnated
Cortés Orihuela et al. (2024)	CHL	I	Rank-rank	Rob.	Top pers. “ remarkable ”
Cartagena (2003)	COL	E	TM	Rob.	Improved then stagnated
Nina et al. (2003)	COL	E	TM	Rob.	Stagnated
Bonilla (2010)	COL	E	β, r, TM	Mix	Improved (sensitive to index)
Galvis & Meisel (2014)	COL	E	β, TM	Mix	Improved (still low)
Dahan & Gaviria (2001)	CRI	E	Sibling ρ	Rob.	Highest mobility in LAC
Jiménez & Jiménez (2019)	ECU	E	OV index	Rob.	Improved (not at bottom)
Binder & Woodruff (2002)	MEX	E	β, r, TM	Mix	Improved (r); TM limited at top
Delajara & Graña (2017)	MEX	E	Rank-rank	Rob.	High persistence, N-S gap
Urbina (2018)	MEX	E	Seq. logit	Rob.	Improved (low); declined (high)
Torche (2020)	MEX	M	β	Aff.	Improved then stagnated
Ciaschi et al. (2025)	MEX	M	$\beta, \text{rank-rank}$	Mix	β fell; rank stagnated
Behrman et al. (2001)	PER	E	β	Aff.	Improved (β drop ~ 0.3)

Continued on next page

Table 7 continued

Study	Ctry	Dom.	Measure	Vuln.	Finding
Benavides & Etesse (2012)	PER	E	TM, log-lin.	Rob.	Absolute up; relative stagnated
Pasquier-Doumer (2002)	PER	E	TM, log-lin.	Rob.	Mixed by region
Gandelman & Robano (2014)	URY	M	SMI, IOV, TM	Mix	Declined
Boado (2010)	URY	C	Log-linear	Rob.	Stagnated (constant flu- idity)
Ortega (2011)	VEN	E	Sibling gaps	Rob.	Non-monotonic
<i>No country-specific study: DOM, SLV, GTM, HND, NIC, PAN, PRY</i>					
<i>Panel B: Cross-country and regional studies</i>					
Hertz et al. (2007)	42 c.	E	β, r	Mix	β fell 50yr; r flat ~ 0.4
Behrman et al. (2001)	5 LAC	E	β , sibling	Mix	Improved (β)
Dahan & Gaviria (2001)	16 LAC	E	Sibling ρ	Rob.	CRI high; COL, MEX, SLV low
Daude & Robano (2015)	18 LAC	E	β, r	Aff.	r tells “ bleaker story ”
Neidhöfer et al. (2018)	18 LAC	E	β, r, ρ, TM	Mix	β halved; r, ρ stable ~ 0.5
Neidhöfer (2019)	18 LAC	E	β	Aff.	Improved
Neidhöfer et al. (2024)	18 LAC	E	β	Aff.	Relative position strengthened
Torche (2014)	LAC	M	Review	—	Class mob. $\neq$ devel- oped world
Torche (2019)	Dev.	E	Review	—	IER $\neq$ IEC due to dis- persion
Van der Weide et al. (2024)	153 c.	E	Multiple	Mix	Ineq-sens. $\downarrow$; ineq-inv. flat

Continued on next page

Table 7 continued

Study	Ctry	Dom.	Measure	Vuln.	Finding
Brunori et al. (2025)	LAC	M	β, r , IEO	Mix	β not pure assoc.; r stable
Muñoz & Jaqué (2024)	94 c.	E	Sibling ρ	Rob.	LAC declining
Ahsan et al. (2023)	53 c.	E	Sibling ρ	Rob.	LAC highest (0.65); declining
Muñoz (2024)	24 LAC	E	Threshold	Rob.	Upward mobility increasing
Jiménez & Jiménez (2019)	18 LAC	E	OV index	Rob.	Improved (not at bottom)
OECD (2018)	OECD+	M	Multiple	Mix	9+ gen. to mean income

Notes on Table 7. **Domain:** E = educational mobility (years of schooling, levels completed); I = income or earnings mobility; C = occupational class mobility (EGP or similar schemes); M = multiple domains in the same study. **Measures:** β = OLS regression coefficient of child on parent outcome (intergenerational regression coefficient, IGRC); r = Pearson correlation; ρ = Spearman rank correlation; TM = transition matrices (cross-tabulation of parent and child categories with upward/downward/immobility rates); SMI = Social Mobility Index ($1 - R^2$ from regression of schooling gap on parental background; Andersen 2009); OV = Overlap index (distance between observed and equality-of-opportunity distributions); IGE = intergenerational elasticity of income (log-log regression); IOV = Index of Opportunity Value; IEO = Inequality of Economic Opportunity (R^2 from regression of outcomes on circumstances); Seq. logit = sequential logit model of educational transitions (Mare 1981); UNIDIFF = uniform difference log-multiplicative layer effect model (Erikson and Goldthorpe 1992); Log-linear = log-linear or log-multiplicative models of association in contingency tables. **Vulnerability:** Aff. (Affected) = the study's primary mobility measure depends on the ratio σ_c/σ_p and will mechanically show improving mobility when educational distributions compress, even if rank persistence is unchanged (Proposition 1); Rob. (Robust) = the measure is invariant to marginal distribution changes; Mix = the study reports both types, permitting comparison.

7.3 Implications for cross-country databases

The Global Database of Intergenerational Mobility (van der Weide et al., 2024) reports $1 - r$ (one minus the Pearson correlation of child and parent years of schooling) as its primary measure of relative mobility for 153 countries. This measure is our Object 1's r component. For the 18 LAC countries where we can compute ρ_{lag} , the GDIM's r tracks our r_A closely but diverges from ρ_{lag} exactly as the four-object framework predicts. The

U-shaped relationship between their $1 - r$ and GDP per capita—high apparent mobility in both the poorest and richest countries—is at least partly an artifact: in the poorest countries, near-degenerate parent distributions attenuate the Pearson correlation toward zero regardless of the actual dependence structure.

7.4 Implications for sibling correlation databases

The recent sibling correlation databases—[Muñoz and Jaqué \(2024\)](#) covering 94 countries, [Ahsan et al. \(2023\)](#) covering 53 developing countries, [Anderson et al. \(2024\)](#) meta-analyzing 18 countries—have substantially expanded the empirical frontier. Propositions 3 and 4 imply that their estimates face two comparability problems that, to our knowledge, have not been addressed.

Comparability over time. The documented decline in sibling correlations in LAC—the headline finding of [Muñoz and Jaqué \(2024\)](#) and [Ahsan et al. \(2023\)](#)—confounds genuine equalization with the fertility transition. As educated families reduced fertility and exited the sibling sample (≥ 2 co-resident children), the between-family variance in the ICC numerator compressed mechanically (Proposition 4). This produces a declining ICC even when the latent family effect σ_a^2 is unchanged. The schooling lag corrects the between-cohort channel in the denominator (Proposition 3) but cannot reach the fertility selection channel in the numerator—only DFL reweighting to a fixed family-composition benchmark can. LAC experienced the fastest fertility decline among developing regions (the average TFR fell from ~ 3.0 in 2000 to ~ 2.0 by 2020), precisely where the documented sibling correlation decline is concentrated. The coincidence is suggestive.

Comparability across countries. The ICC is estimated on the subsample of families with ≥ 2 co-resident children aged 14–25 (or similar windows). This subsample’s composition differs radically across countries. In Chile (90% urban, TFR ≈ 1.7), the sibling sample is a

small, selected fraction of all families—disproportionately poorer and less educated. In Bolivia (40% urban, TFR ≈ 3.5), the sibling sample is nearly representative. The two ICCs are computed on fundamentally different populations. From Proposition 4, a country deep into the fertility transition will have $q(t) \ll p$, mechanically compressing σ_a^2 and deflating the ICC. This generates a spurious positive correlation between a country’s fertility level and its sibling correlation—which could partly account for the cross-country Great Gatsby Curve pattern documented by [Anderson et al. \(2024\)](#). The correction is DFL reweighting to a common reference distribution across countries, analogous to standardizing age distributions before comparing mortality rates. The resulting estimates measure family effects on a comparable population, rather than confounding family effects with household demography.

7.5 Limitations

Three limitations deserve emphasis. First, the Latinobarómetro is a social survey with relatively small samples ($N \approx 500\text{--}750$ per country-wave), generating noisy country-level estimates. Our cross-country conclusions rely on 318 country-wave observations, adequate for trend detection but not for precise country-specific inference.

Second, the parent education question elicits the completed level of the most educated parent. If assortative mating increased over the sample period, this would attenuate the measured decline in ρ and make our “three-quarters artifact” finding conservative.

Third, the within-cohort Gatsby Curve assumes a stationary linear AR(1) process; departures from linearity—threshold effects, subsistence constraints, elite traps—would generate a nonlinear $\omega\text{--}\rho$ relationship. [Nybom and Stuhler \(2024\)](#) show that past structural changes can generate non-monotonic mobility trends across generations; our 25-year window may not capture these dynamics.

8 Conclusion

The literature on intergenerational educational mobility has been stuck between two flawed measures. The intergenerational elasticity confounds positional persistence with distributional compression. The rank-rank regression, designed for continuous income, fails for discrete education because massive ties corrupt both the estimand and the inference.

This paper proposes the *schooling lag*—a respondent’s deviation from her cohort-sex mean schooling—and the *rank-of-lag* estimator as a constructive solution to both problems. Demeaning by cohort strips the between-cohort expansion trend that drives the σ -ratio bias, while simultaneously generating the near-continuity needed for the rank transformation to approximate the true rank correlation. The four-object decomposition framework shows precisely how each transformation—demeaning, ranking, and their composition—peels off a distinct source of bias, nesting the IGE, the demeaned-level slope, the rank-of-levels slope, and the rank-of-lags slope within a unified variance hierarchy. The same logic extends to sibling correlations: the standard measure—the broadest index of family effects in the literature—is mechanically deflated when between-cohort expansion inflates total variance, and the lag-based sibling correlation corrects this (Proposition 3).

The within-cohort Great Gatsby Curve provides the structural interpretation: in a Solon-type model, the schooling lag strips all uniform public investment, leaving only private transmission and luck. The steady-state relationship $\omega^2 = \sigma_\varepsilon^2 / (1 - \rho^2)$ connects within-cohort inequality to rank persistence without any mechanical component, and separates access expansion (which raises means without moving ranks) from quality equalization (which reduces both inequality and persistence).

Applied to twenty-five years of Latinobarómetro data across eighteen Latin American countries, the schooling lag reveals that approximately three-quarters of the observed decline in the population-weighted IGE is a distributional artifact. The within-cohort correlation ρ_{lag} declined modestly—from 0.48 to 0.45 over 25 years—while β_A declined more than three times as fast. The standard IGE overstates progress by a factor of three.

The tide has risen; the rankings have shifted, but only a quarter as much as the headline statistic suggests.

The methodological contribution generalizes beyond Latin America. Any country where intergenerational educational mobility is measured with discrete survey data—the standard setting in all of [Hertz et al.’s \(2007\)](#) forty-two countries, all of [van der Weide et al.’s \(2024\)](#) 153 countries, and the developing-country applications of [Asher et al. \(2024\)](#) and [Ahsan et al. \(2022\)](#)—faces the same σ -ratio contamination that the schooling lag resolves. Recomputing the GDIM’s 153-country estimates using the schooling lag would separate the distributional artifact from the structural signal globally.

The extension to sibling correlations may carry the larger implication. The sibling correlation—the ICC from a random effects model—is widely considered the broadest measure of family effects, and three recent databases ([Muñoz and Jaqué, 2024](#); [Ahsan et al., 2023](#); [Anderson et al., 2024](#)) have made it the workhorse of cross-country and over-time comparisons of equality of opportunity. We show that the ICC faces two mechanical biases that render it incomparable—neither over time nor across countries—without correction.

First, the ICC is incomparable *over time*: educational expansion inflates the denominator (Proposition 3), and the fertility transition compresses the numerator by changing which families contribute sibling pairs (Proposition 4). The two biases are orthogonal—the schooling lag corrects the first but not the second; only DFL reweighting addresses the second. The documented decline in LAC sibling correlations coincides precisely with the region’s fertility transition (TFR from ~ 3.0 to ~ 2.0 between 2000 and 2020), the fastest in the developing world.

Second, the ICC is incomparable *across countries*: countries with different fertility levels, urbanization rates, and household structures contribute systematically different populations to the sibling estimating sample. A country deep into the fertility transition (Chile, TFR ≈ 1.7) estimates the ICC on a small, selected subsample; a high-fertility country (Bolivia, TFR ≈ 3.5) estimates it on a nearly representative one. The cross-country

Great Gatsby Curve for sibling correlations ([Anderson et al., 2024](#)) may partly reflect this compositional confound. The correction is DFL reweighting to a common reference distribution—analogous to standardizing age distributions before comparing mortality rates.

Neither [Muñoz and Jaqué \(2024\)](#) nor [Ahsan et al. \(2023\)](#) corrects for either channel. Applying the schooling lag and DFL reweighting to their databases—94 and 53 countries, respectively—would determine how much of the documented cross-country variation and temporal decline in sibling correlations reflects genuine differences in family effects versus demographic composition. This is a natural and immediate extension of the present paper.

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